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THE ENERGY AND ENVIRONMENTAL FOOTPRINT OF AI

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Artificial intelligence (AI) has the potential to create major economic and social benefits, but also to rapidly escalate electricity demand and its associated environmental impacts. Information availability has been a cornerstone of environmental law for half a century, and this Article argues that providing information to individual, corporate, and other users about the electricity demand and environmental impacts of AI can reduce those impacts without delaying development of the technology. Little is known about how different large language models (LLMs) compare on these metrics, though. To address whether users have access to the information necessary to address this shortcoming, the Article provides the first comparison of the outputs of four AI environmental footprint calculators. The Article finds that inputting the same AI query into all four calculators produces substantial differences in footprint estimates, with one calculator producing an estimate more than 50 times higher than another for the same type of query. These differences suggest that substantial improvements are needed in the disclosure of AI model information, whether through international, national, state, or private standards, to provide reliable estimates of energy use and environmental impacts to users. In turn, more accurate, easily available information can create incentives for reducing the costs, energy demand, and environmental impacts of AI even in a deregulatory era.

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INTRODUCTION

Information is the foundation of environmental governance,[1] yet surprisingly little information is publicly available about the most important and energy-intensive technological development in decades: artificial intelligence (AI). Half of American adults have used Large Language Models (LLMs), with 34% of LLM-users doing so at least once per day.[2] These tools are not just for personal use. In 2025, 88% of businesses reported using generative AI, a broader category that encompasses LLMs and other generative technologies, for at least one business function.[3] AI is even helping to construct the next generation of programming as over 63% of professional programmers reported using generative AI tools for their code in mid-2024.[4]

As AI becomes mainstream, concerns have emerged about its environmental footprint.[5] Data centers, large facilities stocked with state-of-the-art computer chips, execute the tens-of-billions of calculations every second necessary to power the computationally intensive programs underlying AI.[6] They must operate

1. *See, e.g.*, Daniel C. Esty, *Environmental Protection in the Information Age*, 79 N.Y.U. L. REV. 115, 115 (2004) ("Information gaps and uncertainties lie at the heart of many persistent pollution and natural resource management problems."); *see* Cass R. Sunstein, *Informational Regulation and Informational Standing: Akins and Beyond*, 147 U. PA. L. REV. 613, 613–614 (1999) (exploring information disclosure in environmental statutes).

2. LEE RAINIE, CLOSE ENCOUNTERS OF THE AI KIND: THE INCREASINGLY HUMAN-LIKE WAY PEOPLE ARE ENGAGING WITH LANGUAGE MODELS (Imagining the Digit. Future Ctr. ed., 2025), https://imaginingthedigitalfuture.org/reports-and-publications/close-encounters-of-the-ai-kind/close-encounters-of-the-ai-kind-main-report/.

3. Alex Singla et al., *The State of AI in 2025: Agents, Innovation, and Transformation*, MCKINSEY & CO. (Nov. 5, 2025), https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai.

4. *AI: Sentiment and usage*, STACK OVERFLOW (2024), https://survey.stackoverflow.co/2024/ai-sentiment-and-usage-ai-sel-prof.

5. *See, e.g.*, Amy L. Stein, *Artificial Intelligence and Climate Change*, 37 YALE J. ON REG. 890, 917–22 (2020) (discussing the environmental impacts of AI use and proposing ways "to ensure that AI's negative environmental impacts are outweighed by its positive ones"); James O'Donnell & Casey Crownhart, *We Did the Math on AI's Energy Footprint. Here's the Story You Haven't Heard.*, MIT TECH. REV. (May 20, 2025), https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/; Pranshu Verma & Shelly Tan, *A Bottle of Water per Email: the Hidden Environmental Costs of Using AI Chatbots*, WASH. POST (Sept. 18, 2024), https://www.washingtonpost.com/technology/2024/09/18/energy-ai-use-electricity-water-data- centers/; Renée Cho, *AI's Growing Carbon Footprint,* STATE OF THE PLANET (Jun. 9, 2023), https://news.climate.columbia.edu/2023/06/09/ais-growing-carbon-footprint/.

6. Marcus Law, *Google Cloud Unveils Ironwood TPU with 42.5 Exaflops per Pod*, DATACENTRE MAGAZINE (Apr. 9, 2025), https://datacentremagazine.com/articles/google-cloud-next-2025-the-announcements-you-need-to-know (reporting that Google's new Tensor Processing Unit can perform 42.6 exaflops (quintillion floating point calculations per second; a quintillion is a billion billion) and that Nvidia's competing B200 and GB200 processors can produce 15 exaflops per rack). Data centers can be truly massive, reaching millions of square feet and house tens-of-thousands of chips. *See* Sean Farmer, *The Stone in the Cloud: Planning the Resource Demands of Data Centre Industry Through Land Use Law*, 56 U.B.C. L. REV. 435, 435–36 (Nov. 2023) ("[A] data centre is a building that contains equipment running software designed to 'process data requests—to receive, store and deliver—to "serve" data, such as games, music, emails and apps, to clients over a network'."); Alexandra Jonker & Alice Gomstyn, *What is an AI Data*

constantly because power outages can cause data loss and other problems.[7] U.S. data centers consumed 176 terawatt hours (TWh) in 2023,[8] equivalent to the average annual consumption of 17.2 million American households.[9] It is hard to disaggregate AI from other workloads for which data centers are responsible (a report from the International Energy Agency estimates that AI is currently responsible for ~15% of total data center energy), but the continued growth of AI is the key driver for projected increases in overall data center energy demand.[10] Energy consumption from data centers may triple to 580 TWh by 2028,[11] more than the electricity use of Germany, France, or Brazil in 2019.[12] The North American Electric Reliability Corporation, responsible for ensuring a reliable electricity supply across the continent, has warned that growing electricity demand from data centers and increased cooling needs due to climate change could cause rolling blackouts across most of the United States.[13] For instance, the market monitor for PJM Interconnection, the nation's largest electric interconnection, has argued that PJM cannot reliably serve growing data center demand and will face periodic blackouts unless demand growth is curtailed.[14]

Center?, IBM (Feb. 21, 2025), https://www.ibm.com/think/topics/ai-data-center ("An AI data center is a facility that houses the specific IT infrastructure needed to train, deploy and deliver AI applications and services.").

7. Boško Marijan, *Data Center Power Outage*, PHOENIXNAP (Oct. 17, 2025), https://phoenixnap.com/blog/data-center-power-outage.

8. Arman Shehabi et al., *2024 United States Data Center Energy Usage Report*, BERKELEY LAB (2024), https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf.

9. Based on average residential electricity consumption of 10,260kWh per year. There are one billion kWh in one TWh; 176TWh = 1.76x10^11 kWh, and 1.76x10^11/10,260 = 17.2 million. *Electric Sales, Revenue, and Average Price: Table 5.a,* U.S. ENERGY INFO. ADMIN., (Oct. 7, 2025), https://www.eia.gov/electricity/sales_revenue_price/.

10. ENERGY AND AI, INT'L ENERGY AGENCY 56–57 (2025), https://iea.blob.core.windows.net/assets/de9dea13-b07d-42c5-a398-d1b3ae17d866/EnergyandAI.pdf (using electricity consumption from accelerated servers as a proxy for the share of AI in total electricity consumption from data centers).

11. Shehabi et al., *supra* note 8, at 6. It's important to note that energy consumption predictions tend to exceed real energy load growth. Watt-hours (Wh) are a common unit of electricity that measures the power draw (in watts or kilowatts) over time. For example, watching one hour of TV on a 50W television would consume 50Wh.

12. *Electricity Information: Overview – Electricity Consumption*, INT'L ENERGY AGENCY, https://www.iea.org/reports/electricity-information-overview/electricity-consumption (July 2025).

13. *See* 2025 SUMMER RELIABILITY ASSESSMENT, NORTH AM. ELC. RELIABILITY CORP. 7 (2025), https://www.nerc.com/pa/RAPA/ra/Reliability%20Assessments%20DL/NERC_SRA_2025.pdf (finding that most of the US is at high or elevated risk of rolling blackouts due to extreme heat and increased industrial and data mining loads driving load growth).

14. *See* Complaint of Indep. Mkt. Monitor for PJM at 2–3, Indep. Market Monitor for PJM v. PJM Interconnection, L.L.C. No. EL26-30. (FERC Nov. 25, 2025) ("PJM is currently proposing to allow the interconnection of large new data center loads that it cannot serve reliability and that will require load curtailments (black outs) of the data centers or of other customers at times."); *see also* Ethan Howland, *No More PJM Data Centers Unless They Can be Reliably Served: Market Monitor*, UTILITY DIVE (Nov. 26, 2025), https://www.utilitydive.com/news/pjm-data-center-interconnection-market-monitor-ferc-complaint/806527/ ("PJM is considering proposing to allow data center loads that it cannot serve reliably and that will require periodic blackouts for data centers and other customers.").

Generating this electricity emits millions of tons of carbon, including 61 million metric tons (MMT) of CO_2 equivalent (CO_2e) in the United States in 2023 alone.[15] This is equivalent to adding an additional 13 million cars to the road for one year.[16] If current emissions intensities persist, this figure could reach 197 MMT by 2028, equal to about half of total residential emissions in 2022.

AI also drives increases in water usage. Google's data center fleet withdrew as much water in 2024 as 329,635 people (or 51 golf courses) in the U.S..[17] In addition to direct water use, data centers indirectly consume 4.52L/kWh from energy generation, meaning the power supply for even small data centers can require thousands of gallons each year.[18] Although water consumption is an important consideration, it is a complex issue that is outside the scope of this Article.

Despite the concerns about the stress on the electric grid, the environmental impacts of AI, AI's contribution to raising the price of electricity for residential and business users,[19] and pressure on water supplies,[20] federal action is unlikely. A provision banning states from regulating AI for ten years was almost adopted by Congress in 2025, suggesting that comprehensive federal regulatory legislation is

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15. *See* Shehabi et al., *supra* note 8, at 57. Some estimates found that data centers emit 48% more carbon per unit of electricity than the grid, though, which would imply an additional ~30MMT in 2023 and ~100MMT in 2028 barring significant decarbonization. *See* Gianluca Guidi et al., *Environmental Burden of United States Data Centers in the Artificial Intelligence Era*, ARXIV (Nov. 14, 2024), https://arxiv.org/html/2411.09786v1.

16. Based on estimates that the "typical passenger vehicle emits about 4.6 metric tons of carbon dioxide per year." *Greenhouse Gas Emissions from a Typical Passenger Vehicle*, U.S. EPA (Jun. 12, 2025) https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle.

17. ENVIRONMENTAL REPORT, GOOGLE 105 (2025), https://www.gstatic.com/gumdrop/sustainability/google-2025-environmental-report.pdf (reporting that Google data centers withdrew 9.87 billion gallons in 2024). *See also* Cheryl A Dieter & Kristin S Linsey, *Estimated Use of Water in the United States County-Level Data for 2015* [Data set], U.S. GEOLOGICAL SURVEY (2017) https://doi.org/10.5066/F7TB15V5 (reporting that the average American used 82 gallons of water per day for domestic purposes in 2015).

18. Shehabi et al., *supra* note 8 at 57 ("In 2023, energy consumption by a data center translates to national average of 4.52L/kWh of indirect water consumption."). A 1MW data center, tiny compared to modern hyperscale data centers, could indirectly consume up to 39.6 million liters or 10.5 million gallons in a year.

19. *See* Marc Levy, *As Electric Bills Rise, Evidence Mounts That Data Centers Share Blame. States Feel Pressure to Act*, ASSOCIATED PRESS (Aug 9. 2025), https://apnews.com/article/electricity-prices-data-centers-artificial-intelligence-fbf213a915fb574a4f3e5baaa7041c3a ("Monitoring Analytics, the independent market watchdog for the mid-Atlantic grid, produced research in June showing that 70% – or $9.3 billion – of last year's increased electricity cost was the result of data center demand."); Ivan Penn & Karen Weise, *Big Tech's AI Data Centers Are Driving Up Electricity Bills for Everyone*, N.Y. TIMES (Aug. 14, 2025), https://www.nytimes.com/2025/08/14/business/energy-environment/ai-data-centers-electricity-costs.html ("[R]ecent reports expect data centers will require expensive upgrades to the electric grid, a cost that will be shared with residents and smaller businesses through higher rates unless state regulators and lawmakers force tech companies to cover those expenses.").

20. *See, e.g.*, Eli Tan, *Their Water Taps Ran Dry When Meta Built Next Door*, N.Y. TIMES (July 14, 2025), https://www.nytimes.com/2025/07/14/technology/meta-data-center-water.html (discussing the impact of data center construction on a neighboring family's water supply).

unlikely in the near term.[21] Moreover, the White House has adopted executive orders discouraging regulation of AI[22] and has adopted an AI "Action Plan" that expressly directs the Department of Commerce to remove the term "climate change" from an AI risk management standard emerging from the National Institute of Standards and Technology, a public-private standard-setting organization.[23] Some states have adopted regulations regarding AI, but none have required disclosure of AI-driven energy use or environmental impacts.[24] The European Union has regulatory initiatives underway, but it is unclear whether these initiatives will result in energy and environmental disclosures to corporate and household users in the U.S.[25]

Not surprisingly, given the absence of regulation and the risks of transparency (e.g., the potential for activist pressure or perceived misalignment with federal AI goals),[26] AI firms disclose only limited energy and environmental

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21. *See* S. Amend. 2360 to H.R. 1, 119th Cong. § 40012 (2025); *see also* S. Amend. 2814 to S.Amend. 2360 to H.R. 1, 119th Cong. (as passed by Senate, Jul. 1, 2025); Matt Brown & Matt O'Brien, *Senate Pulls AI Regulatory Ban from GOP Bill After Complaints from States*, PBS NEWS (July 1, 2025), https://www.pbs.org/newshour/politics/senate-pulls-ai-regulatory-ban-from-gop-bill-after-complaints-from-states. For reviews of the barriers to public governance over the last several decades, see Michael P. Vandenbergh, *The Emergence of Private Environmental Governance*, 44 ENV'T L. REP. 10125, 10132 fig.1 (2014), and David Uhlmann, *The Quest for a Sustainable Future*, 1 MICH. J. ENV'T & ADMIN. L. 1, 9 (2012).

22. *See* Removing Barriers to American Leadership in Artificial Intelligence, Exec. Order No. 14,179, 90 Fed. Reg. 8741 (Jan. 31, 2025) (ordering the suspension, revision, or rescindment of any agency actions deemed to be at odds with the policy of sustaining and enhancing "America's global AI dominance"); *see also* Accelerating Federal Permitting of Data Center Infrastructure, Exec. Order No. 14,318, 90 Fed. Reg. 35385 (July 28, 2025) (taking steps to "facilitate the rapid and efficient buildout of [AI] infrastructure by easing Federal regulatory burdens"); Promoting the Export of the American AI Technology Stack, Exec. Order No. 14,320, 90 Fed. Reg. 35393 (July 28, 2025) (creating a "coordinated national effort to support the America AI industry by promoting the export of full-stack American AI technology packages").

23. EXEC. OFF. OF THE PRESIDENT, WINNING THE RACE: AMERICA'S AI ACTION PLAN 4 (2025), https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf; *see generally* ELHAM TABASSI ET AL., NAT'L INST. OF STANDARDS & TECH., ARTIFICIAL INTELLIGENCE RISK MANAGEMENT FRAMEWORK (AI RMF 1.0) (2023), http://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf (outlining the new AI risk management standard).

24. *US State-by-State Artificial Intelligence Legislation Snapshot*, BRYAN CAVE LEIGHTON PAISNER LLP, https://www.bclplaw.com/en-US/events-insights-news/us-state-by-state-artificial-intelligence-legislation-snapshot.html (last visited Aug. 13, 2025) (detailing all proposed and enacted State AI legislation).

25. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828, 2024 O.J. (L 1689) 1 (EU), https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng.

26. Some companies may worry that voluntary disclosure will make them a target of environmental activists who view their energy consumption as excessive. *See, e.g.,* GOOGLE'S ECO-FAILURES: AN ENVIRONMENTAL INVESTIGATION INTO ALPHABET INC. 2025*,* KAIROS FELLOWSHIP (2025), https://noclimateresultsfound.com/wp-content/uploads/2025/07/Kairos_NCRFGoogleReport_FINAL.pdf. Others may fear retribution from the federal government—which is responsible for significant investment in AI and data centers—if they are perceived as prioritizing the environment over AI dominance. *See* Matthew F. Ferraro et al., *Draft Executive Order Seeks to Short-Circuit AI State Regulation*, CROWELL & MORING LLP (Nov. 24, 2025), https://www.crowell.com/en/insights/client-alerts/draft-executive-order-seeks-to-short-circuit-ai-state-regulation (detailing a draft executive order that would direct the Department of Justice to sue states that enact "burdensome" AI regulations and regulations

information, and most AI users are unaware of the environmental implications of their AI use.[27] AI advocates argue that the economic benefits of expanding data centers and AI outweigh the environmental impact, and that required disclosure would create greater harms by slowing down the development of beneficial AI applications.[28] In other words, some argue that unleashing AI will be such a benefit for combating environmental threats in the long term that its near-term energy demand and environmental impacts are inconsequential.[29] It is true that AI tools have helped companies improve climate reporting and disclosure,[30] reduce their environmental footprint,[31] and optimize energy generation to increase the use of

"[embedding] DEI in [AI models'] programming." The Order does not specify what may qualify as "burdensome" or "DEI," but it is plausible that environmental considerations would fall under these characterizations given the administration's previous efforts regarding environmental regulation. Although the Order targets states, local governments and private firms are likely to feel similar pressure).

27. *See* Jane E. Miller & Michael P. Vandenbergh, *Reducing the Energy Costs and Environmental Impacts of AI: Understanding User Behavior and the Potential for Information Disclosure* (Vand. Univ. L. Sch. Working Paper 2025) ("Despite high usage, participants' awareness of the resource demands and environmental impacts of generative AI was limited.").

28. *Do the Benefits of Generative AI Outweigh its Environmental Impact?*, CBIZ INSIGHTS (July 25, 2024), https://www.cbiz.com/insights/article/do-the-benefits-of-generative-ai-outweigh-its-environmental-impact; *see also* Ana Pošćić, *The Intersection Between Artificial Intelligence and Sustainability: Challenges and Opportunities*, 8 E.U. & Comp. L. Issues & Chall. Ser. 748, 757 (2024), https://hrcak.srce.hr/ojs/index.php/eclic/article/view/32300 ("AI can improve societal outcome[s] like . . . [alleviating] poverty, [facilitating] quality education, clean water and sanitation, clean energy, sustainable cities, [and] identif[ing] areas of poverty by satellite images.").

29. *See* James Temple, *Sorry, AI Won't "Fix" Climate Change*, MIT TECH. REV. (Sep. 28, 2024), https://www.technologyreview.com/2024/09/28/1104588/sorry-ai-wont-fix-climate-change (describing an essay in which "Sam Altman, the CEO of OpenAI, argued that the accelerating capabilities of AI will usher in an idyllic 'Intelligence Age' unleashing 'unimaginable' prosperity and 'astounding triumphs' like 'fixing the climate'."); Chase DiBenedetto, *Google's Former CEO: AI Advances More Important Than Climate Conservation*, MASHABLE (Oct. 7, 2024), https://mashable.com/article/former-google-ceo-invest-ai-despite-climate-concerns (detailing Ex-Google CEO Eric Schmidt's argument that "current climate goals should be abandoned in favor of a no-bars-held approach to AI investment"); *see generally* Vasudha Sharma & Abheyshek Jamwal, *Artificial Intelligence's Role in Environmental Conservation: A Study on Harnessing Artificial Intelligence for Planetary Preservation*, 7 INT'L J.L. MGMT. & HUMANS. 297 (2024), https://ijlmh.com/paper/artificial-intelligences-role-in-environmental-conservation-a-study-on-harnessing-artificial-intelligence-for-planetary-preservation/ (describing the positive applications of AI in environmental conservation).

30. For examples of AI climate and CO2 reporting tools, see *CO2 AI Product Ecosystem*, CDP WORLDWIDE, https://www.cdp.net/en/insights/co2ai-product- ecosystem (last visited Aug. 11, 2025), *The Future of Climate Management is Here*, PERSEFONI AI, https://www.persefoni.com/persefoniai (last visited Aug. 11, 2025) & *Make Climate Risk Understandable, Measurable, and Actionable*, CLARITY AI, https://clarity.ai/climate/ (last visited Aug. 11, 2025). For a discussion of private governance initiatives directed at environmental problems, see Michael P. Vandenbergh, *Private Environmental Governance*, 99 CORNELL L. REV. 129, 134–37 (2013).

31. *See, e.g.*, *How AI is Helping Amazon Buildings Conserve Water and Improve Energy Efficiency Around the World,* AMAZON, (Mar. 11, 2025), https://www.aboutamazon.com/news/sustainability/amazon-ai-buildings-water-energy-efficiency; Claudia Elena Marinică, *Artificial Intelligence - A Possible Key for Better Results on Tackling Climate Change*, 12 UNION JURISTS ROMANIA L. REV 83, 89–90 (Aug. 8, 2022), https://internationallawreview.universuljuridic.ro/index.php/journal/article/view/7 (discussing the Capgemini Research Institute report "Climate AI: How artificial intelligence can power your climate action

renewables,[32] and that additional benefits are also likely. It is not at all clear, however, that net environmental benefits will arise or that greater disclosure of energy use and environmental impacts will hamper or delay these outcomes.

The lack of disclosure from major AI providers about their energy and water consumption makes it very difficult for experts, policymakers, and the public to understand the effects of AI-driven data center electricity use.[33] Researchers can run experiments using open-source models,[34] but the mechanics of the most popular models are private, and their owners do not report on their environmental footprint.[35] Without data from the source, AI users are left in the dark regarding the energy and environmental footprint of their AI use. Organizations have developed AI energy and environmental footprint calculators to fill some gaps,[36] but they rely on rough estimates and do not have a consistent methodology. These calculators represent an important step in the informational governance of AI, but our analysis suggests that the results are often inconsistent. For instance, the Ecologits calculator estimates that asking Meta's 8 billion parameter[37] Llama 3.1 model to write one page of text (~400 tokens) consumes 1.51Wh.[38] On the other hand, ML.Energy

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strategy" which finds that AI has the potential to help organizations decrease their energy intensity to 11–45% of targets of Paris Agreement).

32. *See* U.S. DEP'T OF ENERGY, *How AI Can Help Clean Energy Meet Growing Electricity Demand* (Aug. 16, 2024), https://www.energy.gov/policy/articles/how-ai-can-help-clean-energy-meet-growing-electricity-demand.

33. O'Donnell & Crownhart, *supra* note 5; *see also* Stein, *supra* note 5, at 920 ("The hope is that increasing transparency and accountability would make researchers put more effort into keeping [ecological] costs low and bring awareness to potential impacts of algorithms.").

34. *See generally* Sasha Luccioni, Bruna Trevelin & Margaret Mitchell, *The Environmental Impacts of AI–Policy Primer*, HUGGING FACE BLOG (2024), https://www.sashaluccioni.com/AI%20+%20Environment%20Primer%20(Hugging Face).pdf (providing an overview of the environmental impacts of the full AI lifecycle, but noting that "the nature and extent of AI's effects are under-documented").

35. Companies like Google and Microsoft report their total environmental footprints in annual reports but do not specify what portion is attributable to AI. Insight into the process can be especially challenging because most data centers handle more than just AI. *See, e.g.*, Dana Kerr, *AI Brings Soaring Emissions for Google and Microsoft, a Major Contributor to Climate Change*, NPR (July 12,2024), https://www.npr.org/2024/07/12/g-s1-9545/ai-brings-soaring-emissions-for-google-and-microsoft-a-major-contributor-to-climate-change ("Google said its greenhouse gas emissions rose last year by 48% since 2019. It attributed that surge to its data center energy consumption and supply chain emissions.").

36. *See, e.g.*, CODECARBON, EcoLogits Calculator, https://huggingface.co/spaces/genai-impact/ecologits-calculator (last visited Aug. 11, 2025); Julien Delavande, *Chat UI Energy: Tracking Energy Use in ChatUI*, HUGGING FACE SPACES, https://jdelavande-chat-ui-energy.hf.space/ (last visited Aug. 12, 2025); DELOITTE, *AI Carbon Footprint Calculator*, https://www.deloitte.com/uk/en/services/consulting/content/ai-carbon-footprint-calculator.html (last visited Aug. 13, 2025); AIX HUMAN, *AI Environmental Footprint*, https://www.aixhuman.co/ (last visited Aug. 13, 2025).

37. Parameters refer to the number of nodes in a LLM algorithm that change as the model is trained (i.e., the number of variables the LLM can modify to better predict the probability of the output tokens). It is often used as a measure of model size or complexity, with more parameters typically associated with more complex or powerful models. *See* Ivan Belcic & Cole Stryker, *What are LLM Parameters?*, IBM, https://www.ibm.com/think/topics/llm-parameters (last visited Apr. 10, 2026).

38. According to figures generated for the original August 2025 analysis included in this Article. EcoLogits has since changed its model coverage and no longer displays estimates for LLaMa 3.1. CODECARBON, *supra* note 36.

Leaderboard estimates that the same prompt on the same model uses just 0.026Wh.[39] A person who uses one footprint calculator will thus come to very different conclusions about the energy costs of AI use than someone who chooses the other. The methodologies of the calculators are publicly available, but without information from LLM providers, the creators must make assumptions and design choices to fill in important gaps without being able to discern which are most plausible. The user, therefore, cannot determine which estimate is more accurate without disclosures from the operators themselves.

Newer generations of LLMs exacerbate challenges in assessing how much energy a prompt uses. Open AI's GPT-5 incorporates multiple different models with different energy usages and environmental impacts, and it automatically chooses which model to use for processing a given query, a process described as "automatic routing."[40] As a result, users have neither control over nor knowledge about which model their query uses and what the energy and environmental impact will be. This practice could be beneficial if routing decisions prioritize resource efficiency, but the algorithm's opacity clouds users' awareness of their environmental impact.

The rapid pace of AI development presents another challenge in monitoring its environmental impacts. For instance, Google reported that the emissions per query for its flagship Gemini model decreased by 98% between May 2024 and May 2025.[41] Research is an intentionally deliberate practice that can take months to produce, in which time legitimate figures about AI may become obsolete. In fact, we ran into this problem in our research for this Arrticle. Between the time of the original analysis conducted for this Article (completed August 2025), the first review (November 2025), and the second review (January 2026) each of the calculators assessed here underwent significant changes. EcoLogits updated its model coverage in Fall 2025, removing most open-source models, thereby limiting comparability with the other calculators, which focus primarily on these open-source models. EcoLogits also made significant changes to its methodology. ChatUI Energy was removed by its creator and is no longer available online. On January 1, 2026, ML.Energy Leaderboard changed its model coverage, its hardware options (i.e., instead of choosing between A100 and H100, users now select between H100 and the newer Nvidia B200 chip), and cover more task types (in addition to significant updates to the interface). AI Energy Score added results from a new round of testing

39. *How Much Energy do GenAI Models Consume? LLM Chatbot Response Generation*, ML ENERGY.LEADERBOARD (May 12, 2025), https://ml.energy/leaderboard/. ML.Energy Leaderboard has since been updated and no longer covers LLaMa 3.1 8B. This figure is based on an estimate pulled directly from the Leaderboard table on Aug. 4, 2025. A token is a numerical representation of four or five characters that allows the model to process data more efficiently. For example, "resting" might be split into [rest] and [ing], corresponding to [123] and [456], while the word "waking" might be split into [wak] and [ing], corresponding to [789] and [456]. *What are tokens and how to count them?*, OPENAI (July 30, 2025), https://help.openai.com/en/articles/4936856-what-are-tokens-and-how-to-count-them.

40. Benji Edwards, *The GPT-5 Rollout has Been a Big Mess*, ARS TECHNICA (Aug. 11, 2025), https://arstechnica.com/information-technology/2025/08/the-gpt-5-rollout-has-been-a-big-mess/.

41. Cooper Elsworth et al., *Measuring the Environmental Impact of Delivering AI at Google Scale*, ARXIV 1 (Aug. 21, 2025), https://arxiv.org/pdf/2508.15734.

(December 2025) and increased model coverage. This Article presents the original analysis from August 2025 both to maintain a snapshot of the landscape from that time and because many of the analytical challenges we faced in August 2025 persist in the recent updates. The updates and the resulting changes in outcomes provide further demonstration of the difficulties users face when attempting to keep up with industry developments without industry disclosures.[42]

This Article begins by examining the ongoing debate about the energy demand of using AI tools. It then examines what we know about AI inference's energy consumption, how the landscape is changing, and how AI energy demand might evolve. It also highlights the variability in estimates by comparing four popular footprint calculators and demonstrating that different assumptions in these footprint calculators result in substantial differences in estimated consumption.[43] The Article concludes with a call for viable public or private governance initiatives that increase transparency and drive progress on energy efficiency and environmental impacts without discouraging or delaying the benefits of AI.

I. DOES INFERENCE MATTER?

AI deployment has two main phases: training and inference. Training is the learning phase, during which the AI model is fed massive amounts of information. Using complex algorithms, the model processes this data to identify patterns and learn connections, much like teaching someone to recognize different animals by showing them countless pictures.[44] Inference is the application phase. During the application phase, the trained model uses the patterns it learned from the processed information to analyze new, unseen data and make predictions, generate outputs, or

42. The initial study calculations, results and additional information about the methodology are on file with the authors. The exact study and the figures in this paper are not replicable. The pace of changes to these calculators and within the AI industry at large make precise estimates of AI energy use exceptionally difficult, and even high-quality estimates are liable to change on short notice. Even if a calculator were to provide a perfect snapshot of per-token or per-query energy consumption––to the extent that a standardized figure for all responses can be accurate at all––it would quickly become outdated. The methodology and analysis of this paper is therefore not intended to posit some "real" energy consumption of the models herein. Instead, the comparisons elucidate how little users know and can know about the energy and environmental footprint of AI, and the discrepancies that seemingly minor differences can introduce. Even if the accurate figures were substantially different than those presented by the calculators, the central finding remains the same: AI users cannot know the real energy and environmental footprint of AI, and researchers are ill equipped to uncover it.

43. Public-facing energy and emissions calculators face similar uncertainties in other fields, too. *See, e.g.,* Zvonimir Rezo et al., *Comparative Review of Carbon Dioxide Emissions Calculators: Airport-Centric Assessment*, 91 TRANSPORTATION RESEARCH PROCEDIA 37 (2025), https://www.sciencedirect.com/science/article/pii/S2352146525006659?via=ihub.

44. *See* Michael Chen, *What Is AI Model Training and Why Is It Important?*, ORACLE CLOUD INFRASTRUCTURE (Dec. 6, 2023), https://www.oracle.com/artificial-intelligence/ai-model-training/ (describing AI model training as an iterative process, in which the system refines its outputs to produce more accurate responses).

perform other tasks.[45] Inference occurs when a user asks an AI tool a question or to perform a task; the AI tool looks through its memory for similar data and predicts an output based on those patterns.[46]

Early in the public deployment of AI, the resource consumption of training took center stage due to its high up-front costs.[47] Some estimated that training ChatGPT 3.0, the model that gained instant notoriety in late 2022, used as much electricity as 1,000 households do in a year.[48] Aided by research into efficient training design and more efficient, AI-focused chips, it may be possible to reduce the energy and environmental effects of AI training,[49] but the growth in model size and frequency of training may yield a net increase in energy demand.[50] It can be

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45. *See* Dave Bergmann, *What is AI Inference?*, IBM, https://www.ibm.com/think/topics/ai-inference (last visited Apr. 10, 2026).

46. New developments of reasoning models include additional steps where the model breaks problems down into smaller steps and "checks its work." *See* Vinayakh, *How Reasoning Models are Transforming Logical AI Thinking*, MICROSOFT DEV. CMTY. BLOG (Feb. 4, 2025), https://techcommunity.microsoft.com/blog/azuredevcommunityblog/how-reasoning-models-are-transforming-logical-ai-thinking/4373194.

47. *See* Alesia Zhuk, *Artificial Intelligence Impact on the Environment: Hidden Ecological Costs and Ethical-Legal Issues*, 1 J. DIGIT. TECHS. & L. 932, 934 (2023) (citing Emma Strubell, Ananya Ganesh & Andrew McCallum, *Energy and Policy Considerations for Deep Learning in NLP*, ARXIV (June 2019)) (finding in 2019 that training a single AI model can emit as much carbon dioxide as the lifetime emissions of five cars).

48. Sarah McQuate, *Q&A: UW Researcher Discusses Just How Much Energy ChatGPT Uses*, UW NEWS (July 27, 2023), https://www.washington.edu/news/2023/07/27/how-much-energy-does-chatgpt-use/. This figure is an estimated one-time cost for training the model, and model training costs can vary significantly. *See* Strubell et. al, *supra* note 47 (noting that in AI models "accuracy improvements depend on the availability of exceptionally large computational resources that necessitate similarly substantial energy consumption").

49. For instance, Chinese startup DeepSeek made headlines when they announced they had trained their V3 model using just 2.8 million GPU hours, far less than their competitors used to train similarly sized models. GPU hours are a good analogue for energy consumption since each GPU draws additional power each second it is being used in training or inference. DEEPSEEK-AI, *DeepSeek-V3 Technical Report,* ARXIV (Dec. 27, 2024), https://arxiv.org/html/2412.19437v1. Meanwhile, Nvidia's new chips are up to 25x more efficient when handling AI workloads than their predecessors. *See* CATALYST W/ SHAYLE KANN: *Can Chip Efficiency Slow AI's Energy Demand?* (Latitude Media, Jul. 18, 2024), https://www.latitudemedia.com/news/catalyst-can-chip-efficiency-slow-ais-energy-demand/.

50. Expectations of "scaling laws" that would govern the energy and computational inputs needed to improve new generations of LLMs have been challenged by recent experience, leading to fears that improvements to model performance and reliability will take far more energy and computational resources than previously expected. *See, e.g.*, Lexin Zhou et al., *Larger and More Instructable Language Models Become Less Reliable,* 634 NATURE 61–68 (2024) (exploring difficulties in refining and scaling models); Melissa Heikkila et al., *Is AI Hitting a Wall?,* FINANCIAL TIMES (Aug. 15, 2025), https://www.ft.com/content/d01290c9-cc92-4c1f-bd70-ac332cd40f94 (examining energy use of models). There is also a movement to introduce continual training, which would likely add significant energy demand. *See* Tongtong Wu et. al., *Continual Learning for Large Language Models: A Survey*, ARXIV (FEB. 7, 2024) https://arxiv.org/pdf/2402.01364 (surveying studies on continual learning, and noting the need for energy-efficient learning models). Furthermore, the demand for rapid improvements and the long time and intensive nature of AI training mean that LLM providers tend to regularly fine tune models and begin working on the subsequent version before or shortly after a release. *See* Jaime Sevilla et. al., *The Longest Training Run*,

reasonably assumed that the energy intensity of training (i.e., the energy needed to train a similarly sized model with similar complexity) will continue to decrease, but the overall direction of energy consumed for AI training is less certain.[51]

Inference's energy and environmental footprint may grow for three main reasons. First, more people will use AI both intentionally (e.g., talking to ChatGPT) and unintentionally (e.g., AI tools embedded in email, video conferencing, and other services).[52] Second, the proliferation of reasoning models, applications that execute additional "thinking" steps before producing the output, will increase the complexity and energy demand of LLMs.[53] Third, the deployment of agents (i.e., AI applications that interface with other tools, often tools connected to the internet, to accomplish a specific task) and other AI-empowered systems that continually operate will increase the volume and regularity of inference.[54]

Access to LLM inference is currently priced below the cost of delivering the service, meaning providers are choosing to lose billions of dollars in order to build a customer base.[55] At some point, AI providers will need to generate profits, and a steep increase in the cost of inference to retail and corporate consumers may substantially reduce demand. Some industry analysts worry that consumers' demand curves are incompatible with the revenue necessary to keep these companies

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EPOCH AI (AUG. 17, 2022) https://epoch.ai/data-insights/longest-training-run (arguing that long training runs are inefficient).

51. There is a clear trend of decreasing energy intensity of computing operations (measured as the energy consumed to perform a fixed number of computations, such as Wh/FLOP), but the increasing complexity of models and diminishing returns in training successive generations could very well lead to growing energy and environmental costs of training, despite growing efficiency in the computing hardware. *See* Jared Fernandez et al., *Efficient Hardware Scaling and Diminishing Returns in Large-Scale Training of Language Models*, TRANS. MACH. LEARNING RES. (2025) ("Inefficient scaling and parallelization methods will worsen the energy efficiency and environmental cost of model training, as architectures and hardware platforms grow in scale.").

52. *See* U.N. Conference on Trade and Development, 2025 Technology and Innovation Report: Inclusive Artificial Intelligence for Development, 8 U.N. Doc. UNCTAD/TIR/2025 (07 Apr. 2025) ("Since 2022, there has been . . . a surge in interest in Generative AI . . . with organizations across different countries and industries experimenting with its use in a wide range of tasks, including content creation, product development, automated coding, and personalized customer service.").

53. *See* Maximilian Dauner & Gudrun Socher, *Energy Costs of Communicating with AI*, FRONTIERS COMMUN. 1, 5 (Jun. 19, 2025), https://www.frontiersin.org/journals/communication/articles/10.3389/fcomm.2025.1572947/full ("From an environmental perspective, reasoning models consistently exhibited higher emissions, driven primarily by their elevated token production.").

54. *See* Patrick K. Lin, *The Cost of Training a Machine: Lighting the Way for a Climate-Aware Policy Framework That Addresses Artificial Intelligence's Carbon Footprint Problem*, 34 FORDHAM ENV'T L. REV. 1, 18 (Spring 2023) ("In contrast, even though training typically involves repetition, inference is performed every single day, nonstop, for as long as the AI system or device is in use.").

55. *See, e.g.*, Ashley Capoot, *OpenAI's Altman is Still Looking to Spend After GPT-5 Launch and is 'Willing to Run the Loss'*, CNBC NEWS (Aug. 8, 2025), https://www.cnbc.com/2025/08/08/chatgpt-gpt-5-openai-altman-loss.html ("[In 2024] OpenAI expected about $5 billion in losses on $3.7 billion in revenue. OpenAI's annual recurring revenue is now on track to pass $20 billion [in 2025], but the company is still losing money.").

solvent.[56] Thus, considerable uncertainty exists about demand for AI inference after the price to consumers reflects the actual cost of providing the services.

New developments in technology and AI software can be expected to increase the efficiency of AI models and reduce energy demand, but these developments will occur alongside the rapid expansion of AI use. The long-term net effect of model training and inference on these countervailing trends is unclear, but the near-term result is greater pressure on the electric grid and the environment, especially if contemporary estimates are used to justify massive new investments in data centers and computational power. Three main perspectives have emerged in the debate about the environmental footprint of AI inference: (1) inference is environmentally irrelevant; (2) the value of inference outweighs its environmental costs; and (3) inference poses substantial environmental risks.

A. Inference is Environmentally Irrelevant

Individuals who hold the opinion that inference does not raise important environmental concerns point to modest estimates of energy usage, such as the widely-cited assertion that ChatGPT uses 3Wh per query, or OpenAI CEO Sam Altman's more recent assertion that an average ChatGPT interaction uses 0.34Wh.[57] Individuals with this perspective highlight that actions that typically do not garner much attention for their environmental footprint, like using a microwave or TV, use as much or more energy.[58] Those with this perspective also rely on this type of comparison to argue that AI inference is a drop in the bucket and that attention should be focused elsewhere.[59] As some models migrate from centralized data centers to individuals' devices, this comparison may grow more salient because AI energy demand will be embodied in charging one's computer or phone – something most people do every day already.[60]

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56. *See, e.g.*, Edward Zitron, *AI is a Money Trap*, WHERE'S YOUR ED AT (Aug. 6, 2025), https://www.wheresyoured.at/ai-is-a-money-trap/ (arguing consumers would be unwilling to pay for AI services if companies charged in line with their actual costs).

57. Sasha Luccioni et al., *Misinformation by Omission: The Need for More Environmental Transparency in AI*, ARXIV (Jun. 18, 2025), https://arxiv.org/pdf/2506.15572 (noting that 53% of media articles cite 3Wh per ChatGPT query, despite the figure being misleading); Sam Altman, *The Gentle Singularity*, SAM ALTMAN BLOG (Jun. 10, 2025), https://blog.samaltman.com/the-gentle-singularity.

58. *See, e.g.*, Andy Masley, *Using ChatGPT is Not Bad for the Environment*, SUBSTACK (Jan. 13, 2025), https://andymasley.substack.com/p/individual-ai-use-is-not-bad-for (outlining everyday actions that consume similar amounts of energy to an AI prompt); Josh You, *How Much Energy Does ChatGPT Use?*, EPOCH.AI (Feb. 7, 2025), https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use; Hannah Ritchie, *What's the Carbon Footprint of Using ChatGPT?*, SUBSTACK (May 5, 2025), https://substack.com/@hannahritchie/p-162892345.

59. *See, e.g.*, Masley, *supra* note 58; Altman, *supra* note 57; Marcel Salathé, *Does ChatGPT use 10x More Energy than a Standard Google Search?*, ENGINEERING PROMPTS (Jan. 16, 2025), https://engineeringprompts.substack.com/p/does-chatgpt-use-10x-more-energy.

60. At present, the computational intensity, complexity, and knowhow needed to run a model locally keep most on the cloud (especially the "best" models), but AI providers are making a concerted effort to enable local operation of LLMs. *See* Jay Stanley, *What's the Future of AI Language Models as a*

Critics of this perspective, on the other hand, argue that existing energy estimates are unsubstantiated, and that the scale of AI adoption will make even small contributions accumulate into a major environmental challenge.[61] Both the 3Wh and 0.34Wh estimates come from sources with unclear reliability (whether from bias or uncertainty) and, without disclosures from AI operators, it is impossible to know exactly how much energy each inference uses. Furthermore, it is hard to explain the projections that AI will cause a major spike in U.S. energy demand in the coming years without also accepting the implication that inference will consume substantially more energy in the future since inference is responsible for up to 90% of the lifecycle energy of an AI model.[62] For example, if AI energy use is as marginal as figures like Altman suggest, the recommendation made by OpenAI to add 100GWs[63] – just under 10% of the total energy capacity in the US as of 2023[64] – of new energy capacity each year to achieve "AI dominance" would seem nonsensical. Models that can run locally, meanwhile, do not necessarily consume less energy; although limiting transmission of data to and from centralized facilities,[65] transitioning to smaller

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Decentralized Technology?, ACLU (Aug. 25, 2025) https://www.aclu.org/news/privacy-technology/decentralized-llms. For instance, OpenAI released its first open-source model intended to run locally on consumer hardware (i.e., powerful laptops or home computers) in August 2025. *Introducing gpt-oss*, OPENAI (Aug. 5, 2025) https://openai.com/index/introducing-gpt-oss/.

61. *See, e.g.*, Eshta Bhardwaj, Rohan Alexander & Christoph Becker, *Limits to AI Growth: The Ecological and Social Consequences of Scaling*, ARXIV (Jan. 29, 2025), https://arxiv.org/html/2501.17980v1; Christian Bogmans et al., *Power Hungry: How AI Will Drive Energy Demand,* IMF: WORKING PAPERS (Aug. 21, 2025), https://www.imf.org/en/Publications/WP/Issues/2025/04/21/Power-Hungry-How-AI-Will-Drive-Energy-Demand-566304 (projecting increased energy prices and emissions based on AI-driven data center growth); Kevin M. Stack & Michael P. Vandenbergh, *The One Percent Problem*, COLUM. L. REV. 111, 1388 (2011), https://scholarship.law.vanderbilt.edu/faculty-publications/226/ ("Our concern is . . . when the one percent argument is made in circumstances where small contributors account for so much of a regulatory problem that the social goal cannot be met without regulating many one percent sources."); Michael P. Vandenbergh, *The Individual as Polluter*, ENV'T L. REP. (Nov. 2005) (arguing for the treatment of individual behavior as a discrete source of pollution).

62. *See* Shehabi et al., *supra* note 8 (noting that from 2018 to 2023, U.S. data center use grew from 1.9% to 4.4% of total U.S. electricity consumption); INT'L ENERGY AGENCY, *supra* note 10 at 63 ("From 2024 to 2030, data centre electricity consumption grows by around 15% per year."); Radosvet Desislavov, Fernando Martínez-Plumed & José Hernández-Orallo, *Trends in AI Inference Energy Consumption: Beyond the Performance-vs-Parameter Laws of Deep Learning*, 38 SUSTAINABLE COMPUTING: INFORMATICS AND SYSTEMS 100857 (Apr. 2023) ("inference accounts for up to 90% of the machine learning costs for deployed AI systems.").

63. Letter from Christopher Lehane, Chief Global Affairs Officer, Open AI, to Michael Kratsios, Office of Science and Technology Policy (Oct. 27, 2025), https://cdn.openai.com/pdf/21b88bb5-10a3-4566-919d-f9a6b9c3e632/openai-ostp-rfi-oct-27-2025.pdf.

64. *See Electricity Generation, Capacity, and Sales in the United States*, U.S. ENERGY INFO. ADMIN. (Apr. 4, 2024), https://www.eia.gov/energyexplained/electricity/electricity-in-the-us-generation-capacity-and-sales.php (the US had roughly 1,189GW of electricity capacity at the end of 2023).

65. Siavash Alamouti, *Quantifying Energy and Cost Benefits of Hybrid Edge Cloud: Analysis of Traditional and Agentic Workloads,* ARXIV (2025), https://arxiv.org/html/2501.14823v2 (finding hybrid edge cloud environments "[achieve] energy savings of up to 75%" compared to a centralized cloud. Hybrid edge cloud environments "can process tasks locally on end devices when possible and offloads only high-resource tasks to the cloud.").

models capable of running on-device,[66] and decreased idle time[67] may reduce energy consumption, existing consumer hardware capable of running AI loads remains less energy efficient than its commercial counterpart.[68] Proponents of the view that AI inference is environmentally insignificant correctly point out that inference currently makes up a relatively small portion of overall electricity consumption. It is not clear, however, that these comparisons make the energy use of inference an unimportant discussion.[69] At this point, a wide range of factors may affect the importance of AI inference for energy demand and environmental impacts, but it is clear that inference may play an important role.

66. Tanushri Kadam & Safi Ansari, *A Comparative Study of Energy Efficiency in Cloud and Edge Computing Paradigms,* 10 INT'L J. SCI. DEV. & RSCH. b621 (2025), https://ijsdr.org/viewpaperforall.php?paper=IJSDR2505177 (finding that edge computing reduces energy consumption in smaller or more intermittent tasks compared to executing these tasks on a centralized cloud. This implies that models capable of running locally would incur energy savings).

67. Since local LLMs are more likely to handle intermittent requests, consumer devices can execute tasks in short bursts then return to low-energy states. Meanwhile, centralized cloud servers use significant amounts of energy to keep servers in a ready state when they are idle. Mufakir Qamar Ansari & Mudabir Qamar Ansari, *Racing to Idle: Energy Efficiency of Matrix Multiplication on Heterogeneous CPU and GPU Architectures*, ARXIV (Jul. 26, 2025), https://arxiv.org/abs/2507.20063 (highlighting how the tendency for consumer electronics to handle tasks in short bursts can save up to 98% of energy consumption); Yiran Lei et al., *The Energy Cost of Execution-Idle in GPU Clusters*, ARXIV (Apr. 6, 2026), https://arxiv.org/html/2604.04745v1 ("[GPUs] can remain at high power even when visible activity is near zero. We call this state *execution-idle*. . . . [I]t accounts for 19.7% of in-execution time and 10.7% of energy.").

68. The most recent top-of-the-line Nvidia options are the RTX 5090 (for consumers) and B200 (for data centers). One RTX 5090 needs system power of 1kW and can perform 3,352 trillion operations per second at the precision that is standard for modern AI workloads. Meanwhile, one rack of B200s (10 units), which *are* designed specifically for AI, has a peak power demand of ~14.3kW and can perform 144,000 trillion FP4 operations per second, meaning it can do about 43 times as many operations per second while consuming 14 times as much energy. If given the same load, it is probable that the consumer device would use more energy than the more powerful data center architecture. The efficiency of on-device and cloud computing depends on many variables beyond this basic comparison, though. *See Compare GeForce Graphics Cards,* NVIDIA, https://www.nvidia.com/en-us/geforce/graphics-cards/compare (last visited Apr. 9, 2026); *NVIDIA DGX B200*, NVIDIA, https://www.nvidia.com/en-us/data-center/dgx-b200 (last visited Apr. 9, 2026). Nvidia announced a newer generation in March 2026 that claims to achieve up to 35 times as much inference throughput (i.e., tokens per second) per watt as the B200, though official specifications including max power draw are not yet available. *See NVIDIA Vera Rubin Opens Agentic AI Frontier*, NVIDIA, http://nvidianews.nvidia.com/news/nvidia-vera-rubin-platform (last visited Apr. 13, 2026).

69. *See* Stack & Vandenbergh, *supra* note 61 (explaining how the fact that individual organizations, industries, and countries make up a small portion of total emissions is a barrier to regulation). For instance, the fact that residential lighting makes up just 1.68% of electricity consumption in the United States can undercut support for the value of residential lighting efficiency upgrades despite the large total emissions reductions arising from those upgrades. *See Use of Electricity*, U.S. ENERGY INFO. ADMIN. (Dec. 18, 2023), https://www.eia.gov/energyexplained/electricity/use-of-electricity.php.

B. The Value of Inference Outweighs its Environmental Costs

Another perspective asserts that disclosure and regulation should not occur because the utility of AI outweighs even a significant environmental footprint.[70] Inference is responsible for up to 90% of deployed AI tools' energy footprints, but little is known about its marginal costs.[71] AI leaders like Google's Eric Schmidt and OpenAI's Sam Altman have suggested that unleashing AI will enable us to solve major societal problems including climate change and environmental degradation.[72] This perspective may prove to be true, but it places all of society's chips on a single high-stakes gamble that AI will invent ways to rapidly and inexpensively remove greenhouse gases from the atmosphere and alleviate other environmental harm, such as groundwater depletion. It also contradicts the many estimates that AI will drive a long-term surge in energy demand, much of which will be met with new natural gas-fueled plants.[73]

Those with slightly less optimistic projections may point to AI's ability to improve efficiency or accelerate research into greener technologies as a net gain for the environment.[74] For example, if AI can meaningfully decrease the amount of time programmers have their machines on, are working in air conditioned offices, or are brewing coffee to meet a crunch, it could result in a net decrease in resource consumption, even if the AI tools are resource-intensive.[75] Moreover, proponents argue that if AI makes previously impossible research achievable, it may generate new insights that help us combat environmental problems. This perspective still fails to address the apparent market reaction: if this is more likely than not, why are projections suggesting that major increases will occur in net electricity demand,

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70. This will overlap with the first group, since many who believe inference's footprint is minimal are also likely to believe that the benefits of AI are greater than those costs. *See, e.g.*, Newal Chaudhary, *AI and the Fight Against Climate Change: Opportunities and Challenges for Environmental Law*, 6 INT'L J.L. MGMT. & HUMANITIES 390, 391–94 (2023) (describing ways in which AI may be able to help solve climate change and environmental degradation).

71. *See* Desislavov, Martínez-Plumed & Hernández-Orallo, *supra* note 62, at 7 (noting limitations in conducting research on AI energy consumption due to incomplete information, particularly regarding inference costs).

72. DiBenedetto, *supra* note 29; Temple, *supra* note 29; *see also* Lin, *supra* note 54, at 5 ("A 2018 survey conducted by Intel and the research firm Concentrix found that 74 percent of business-decision makers working in environmental sustainability agree that Al will help solve long-standing environmental challenges.").

73. Shehabi et al., *supra* note 8, at 5; INT'L ENERGY AGENCY, *supra* note 10, at 49 ("Natural gas generation to meet data centre demand increases by around 175TWh from today's level to 2035 in the Base Case, mostly concentrated in the United States.").

74. *See, e.g.*, Nicholas Stern et al., *Green and Intelligent: The Role of AI in the Climate Transition*, 4 NPJ CLIMATE ACTION 1 (2025), https://www.nature.com/articles/s44168-025-00252-3 (identifying key areas where AI applications can contribute to emissions reductions).

75. *See* Jules White, *How Much Energy Does AI Use Compared to Coffee?*, LINKEDIN, https://www.linkedin.com/posts/jules-white-5717655_gpt-4-prompt-energy-vs-keurig-coffee-brew-activity-7333158270011949058-JFbE/ (last visited Apr. 9, 2026) (noting that the energy it takes to brew on cup of coffee has the same energy demand as 100 prompts).

causing utilities to make major investments in new gas-fired electricity generation and increasing stress on corporate climate commitments?[76]

Both positions maintain that the output of AI is more valuable than the resources that go into powering it. Advocates of this net-gain perspective are likely to promote expanding AI since they see the potential gains as greater than the costs. It remains important to consider that the way in which AI services are powered can change the analysis (e.g., coal-powered data centers will have different social consequences than solar with storage or nuclear power).[77] Within this view, reducing or shifting the timing of energy demand can increase the margin between AI's positive and negative effects. These advocates also contend that rapid hardware and software upgrades will minimize future environmental costs.[78] The argument that AI will produce net social and environmental benefits is not in tension with proposals for enhanced disclosure or energy-saving initiatives, so long as disclosure does not substantially delay or increase the costs of AI.

In summary, some AI advocates argue that the alternatives to AI are more environmentally harmful than AI, that disclosing and reducing emissions will slow the development of AI, and that slowing its development is more harmful overall given the possible positive implications of the technology.

C. Inference Poses Substantial Environmental Risks[79]

Finally, some argue that AI inference is a serious threat to environmental goals. Advocates of this viewpoint believe that increased energy demand from the data centers serving AI have led to major increases in projected energy demand, and meeting this demand may conflict with carbon emissions goals.[80] Some assert that

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76. For a discussion of how projections tend to overestimate future energy demand, see *infra* at note 90.

77. *See* Eamon Farhat, *AI Data Center Growth Means More Coal and Gas Plants, IEA Says*, BLOOMBERG (April 9, 2025), https://www.bloomberg.com/news/articles/2025-04-10/ai-data-center-growth-means-more-coal-and-gas-plants-iea-says (noting renewable energy is not sufficient to meet increased energy demand from data centers). Some utilities are already calling for more fossil fuel-based power production to meet projected demand. *See, e.g.,* Mason Adams, *Georgia Power Asks to Keep Coal Plants Burning to Meet AI Demand*, SE ENERGY NEWS (Feb. 3, 2025), https://www.canarymedia.com/newsletters/georgia-power-asks-to-keep-coal-plants-burning-to-meet-ai-demand.

78. Proponents may argue that the Jevons Paradox suggests otherwise. *See* Greg Rosalsky, *Why the AI World is Suddenly Obsessed with a 160-Year-Old Economics Paradox*, PLANET MONEY NEWSL. (Feb. 4, 2025), https://www.npr.org/sections/planet-money/2025/02/04/g-s1-46018/ai-deepseek-economics-jevons-paradox (explaining Jevons paradox, which finds that increases in efficiency can lead to rapid increases in demand, resulting in net-positive consumption). This also largely ignores the embodied environmental costs of buying thousands of new chips every couple of years.

79. Many people who hold this position would also point to water and critical mineral consumption. These are essential parts of the discussion, but this piece focuses on the energy demand of AI.

80. *See, e.g.,* DATA CRUNCH: HOW THE AI BOOM THREATENS TO ENTRENCH FOSSIL FUELS AND COMPROMISE CLIMATE GOALS, CTR. FOR BIOLOGICAL DIVERSITY (2025); Tianqi Xiao et al., *Environmental Impact and Net-Zero Pathways for Sustainable Artificial Intelligence Servers in the USA*, 8 NATURE SUSTAINABILITY 1541, 1541 (2025), https://doi.org/10.1038/s41893-025-01681-y ("[T]he

the aggregate effect of inference is large, even if any individual use is small.[81] Since AI needs data centers and data centers need firm energy, fossil fuels may make up a larger share of data centers' electricity supply than they do of the rest of the grid's electricity supply.[82] This means that AI infrastructure in the US alone uses as much energy and emits as much carbon as many countries.[83]

The challenges of AI inference energy demand and corresponding firm energy supply are likely to grow in the coming years. Projections that data center energy demand could triple by 2028 suggest that emissions from electricity use would almost certainly increase.[84] The projected load growth from increased AI usage is also being used to justify adding new fossil fuel infrastructure and bringing older facilities back online.[85] More use also means that processing units (advanced computer chips necessary for AI, like GPUs or TPUs) need to be replaced more frequently, resulting in additional manufacturing emissions.[86]

It is plausible that current trends will not continue in the future and that the environmental footprint of AI will rapidly diminish as the technology matures. DeepSeek's low training costs and the high performance of relatively efficient new models like GPT-4.1nano and LLaMa 3.3 70B suggest that technological breakthroughs might be able to flatten the curve of AI energy use.[87] Similarly, the

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deployment of AI servers across the United States could generate . . . additional annual carbon emissions from 24 to 44 Mt CO2-equivalent between 2024 and 2030"); GOOGLE, *supra* note 17, at 105 (showing that Google's emissions have increased nearly 50% since 2021, which poses challenges for their net-zero target); MICROSOFT CORP., 2025 ENVIRONMENTAL SUSTAINABILITY REPORT: ACCELERATING PROGRESS TO 2030 12 (2025), https://cdn-dynmedia-1.microsoft.com/is/content/microsoftcorp/microsoft/msc/documents/presentations/CSR/2025-Microsoft-Environmental-Sustainability-Report.pdf (finding that Microsoft's emissions have increased 23.4% relative to 2020 baseline).

81. *See, e.g.*, Nidhal Jegham et al., *How Hungry is AI? Benchmarking Energy, Water, and Carbon Footprint of LLM Inference*, ARXIV (May 15, 2025), https://arxiv.org/html/2505.09598v2 (noting inference can account for up to 90% of a model's total lifecycle).

82. *See* Guidi et al., *supra* note 15, at 6–10 (finding that fossil fuel power plants accounted for 56% of the electricity consumed by data centers, resulting in an average carbon intensity 48% higher than the national average). "Firm energy" refers to sources that generate a consistent amount of energy over an indefinite period of time.

83. *See* Guidi et al., *supra* note 15, at 7 (finding US data centers produced 105 million tons CO2e in 2024); this amounts to more total emissions than Peru, New Zealand, or Kenya in 2024. *See* Monica Crippa et al., *GHG Emissions of All World Countries – 2025 Report*, PUBL'N OFF. EUR. UNION at 215, 202, 170 (2024), https://data.europa.eu/doi/10.2760/9816914.

84. Shehabi et al., *supra* note 8, at 11; *see* Stein, *supra* note 5, at 917 (noting that AI data centers are projected to consume between 8% (best case) and 21% (expected) of global electricity in 2025).

85. *See* Matteo Wong, *The False AI Energy Crisis*, THE ATLANTIC (Feb. 11, 2025), https://www.theatlantic.com/technology/archive/2025/02/ai-energy-crisis-fossil-fuels/681653/.

86. It is possible that replacing chips more regularly will result in less energy consumption, but the net effect is dependent on the relative efficiency of the original and replacement chips, as well as the embedded energy or carbon footprint of manufacturing those chips. Further, the Jevons Paradox suggests that improved energy efficiency of new-generation computing chips could drive increased energy demand for both training and inferences. *See generally*, *infra* notes 181, 182, and 184.

87. *See* Peter Hanbury et al., *DeepSeek: A Game Changer in AI Efficiency?*, BAIN & CO., https://www.bain.com/insights/deepseek-a-game-changer-in-ai-efficiency/ (last visited Feb. 8, 2026). ("The company claims to have trained [DeepSeek] for just $6 million . . . vs. the $80 million to $100 million cost

effort to electrify many products, from cars to heaters, will likely induce significantly more load growth than AI itself,[88] meaning that attention must also be given to other sources of new electricity demand.

With that said, it is still hard to argue that AI inference is not environmentally consequential based on current evidence and trends. Even if AI ultimately reduces net energy consumption, the immediate consequence is huge investments in highly consumptive data centers and a new reliance on fossil fuel generation. Even if those who assert that inference is not environmentally consequential are correct, current projections are driving investment decisions that are environmentally consequential.[89] Meanwhile, greater transparency from AI providers surrounding their energy consumption will not negatively affect the performance of models or the significant potential for environmental benefits, so it is unclear why individuals who project AI as a net positive technology would not also support disclosure.

One way to simplify this debate is to induce transparent disclosure from AI providers without adopting regulations that are likely to delay the development of the technology and its environmental benefits. It is possible that AI will follow its predecessors by falling well short of expected energy demand;[90] however, investing in fossil fuel infrastructure locks countries into decades of reliance on those fuels even if the projected AI energy demand never materializes. Clear disclosure is one option to reduce the risk of locking in capital investments that will become outdated and

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of GPT-4."); Jegham et al., *supra* note 81, at 8 (noting the low energy consumption for GPT 4.1nano and LlaMa-3.3 70B models relative to other AI models).

88. *See* ELLA ZHOU & TRIEU MAI, ELECTRIFICATIONS FUTURES STUDY: OPERATIONAL ANALYSIS OF U.S. POWER SYSTEMS WITH INCREASED ELECTRIFICATION AND DEMAND-SIDE FLEXIBILITY 3 (2021), https://docs.nrel.gov/docs/fy21osti/79094.pdf (finding that in a high electrification scenario, electricity consumption in 2050 could increase by as much as 81% relative to 2018).

89. *See, e.g.*, Zachary Skidmore, *Welcome to Gas Land—How Natural Gas is Powering the US AI Boom*, DATA CENTER DYNAMICS (May 1, 2025), https://www.datacenterdynamics.com/en/analysis/welcome-to-gas-land-how-natural-gas-is-powering-the-us-ai-boom/ ("[U]tilities facing massive data center pipeline capacity increases are turning to the natural gas sector to meet forecaster demand."); Diana DiGangi, *Trump Unveils $92B in Energy and AI Investments for Pennsylvania*, UTILITYDIVE (Jul. 17, 2025), https://www.utilitydive.com/news/trump-pennsylvania-investments-energy-ai-blackstone-google/753287/ (reporting a $92 billion investment in energy and AI investments in Pennsylvania, part of which will go towards building and operating new natural gas-based power plants); *see also* Declaring a National Energy Emergency, EXEC. OFF. OF THE PRESIDENT (Jan. 20, 2025), https://www.whitehouse.gov/presidential-actions/2025/01/declaring-a-national-energy-emergency/ (defining "energy resources" as "crude oil, natural gas, lease condensates, natural gas liquids, refined petroleum products, uranium, coal, biofuels, geothermal heat, the kinetic movement of flowing water, and critical minerals."). The White House's definition of energy resources will almost certainly affect the energy sources prioritized through the AI Action Plan. *See supra* note 26.

90. *See* Juan Pablo Carvallo et al., *Long Term Load Forecasting Accuracy in Electric Utility Integrated Resource Planning*, 119 ENERGY POLICY 410 (Aug. 2018) (noting that utilities forecasts of load grow continually over-estimate actual future load, even after downward revisions to forecasts in integrated resource planning documentation); Michael Wara, Danny Cullenward & Rachel Teitelbaum, *Peak Electricity and the Clean Power Plan*, 28 ELECTRICITY J. (May 2015) (arguing that the electricity sales projections underlying the EPA's Clean Power Plan "will mostly likely overestimate true demand for power . . . based on historical forecast error experience").

inconsistent with environmental goals without delaying the development of the technology.

II. How Large is AI Energy Demand?

In the absence of consistent reporting from AI developers and data center operators, researchers have conducted independent studies estimating the energy demand of AI inference. This section highlights estimates of AI energy consumption in the AI literature and from public-facing footprint calculators. We find that estimates tend to follow certain patterns (e.g. model size is correlated to energy consumption) but also exhibit significant variation in their estimates of energy consumption.

The lone lifecycle assessment of a major AI model's inference comes from French startup Mistral AI, whose Large 2 model emits 1.14g CO2 equivalent for a 400-token response (roughly one page of generated text).[91] Although Mistral AI does not disclose energy consumption (which is the metric used by many studies to control for variation in energy intensity of electricity supply), we can infer based on the average emissions intensity of the U.S. grid (0.367g CO2e per Wh)[92] that the Mistral AI Large 2 model uses approximately 3.11Wh for 400-tokens.[93] This may serve as a valuable benchmark, but it is difficult to extrapolate it to other models due to differences in model architecture, size, data center hardware, and other factors. This estimate does fall within the range suggested by the literature, though.

More recently, Google released its own assessment of the marginal electricity consumption and emissions of the median prompt on its Gemini model, reporting that the median prompt consumes 0.24Wh and emits 0.03gCO_2e.[94] Without disclosure of the characteristics of this median prompt, however, it is impossible to benchmark Google's estimate (i.e., we cannot know the per unit costs or whether, for instance, a relatively small number of power users are consuming

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91. *Our contribution to a global environmental standard for AI*, MISTRAL AI (Jul. 22, 2025), https://mistral.ai/news/our-contribution-to-a-global-environmental-standard-for-ai. Mistral's lifecycle assessment includes the impact of inference, amortized training costs, and supply chain emissions. *Id*. As we noted above, one token usually corresponds to four characters and represents a single unit of data that the model takes in or produces. *See* OPENAI, *supra* note 39.

92. *How Much Carbon Dioxide is Produced per Kilowatt-Hour of U.S. Electricity Generation?*, U.S. ENERGY INFO. ADMIN. (Dec. 11, 2024), https://www.eia.gov/tools/faqs/faq.php?id=74&t=11 (showing the emissions intensity of U.S. grid).

93. There is reason to believe Mistral used the U.S. grid's average since the benchmark is against time streaming video in the U.S. To calculate the energy consumption based on emissions intensity of the grid and total emissions, we divided the total emissions (1.14g CO2e) by the emissions intensity (0.367g CO2e/Wh or 367g CO2e/kWh) and arrive at 3.11Wh. If Mistral were to use the French emissions intensity, Large 2 would consume 26.51Wh per 400 tokens because the French grid only emits 43g CO2e per kWh (0.043g CO2e/Wh) so reaching 1.14g CO2e would require greater energy consumption. *See Greenhouse Gas Emission Intensity of Electricity Generation in Europe*, EUR. ENV'T AGENCY (Jun. 27, 2025), https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emission-intensity-of-1 (showing the emissions intensity of European countries).

94. Elsworth et al., *supra* note 41, at 7.

significantly more energy per prompt). Google's effort to disclose this information is laudable and puts Google ahead of many of its competitors, but the information disclosed is insufficient to truly evaluate the environmental footprint of Gemini.

A. The Literature

Estimates of the energy and environmental footprint at all stages of AI development and deployment suffer from substantial uncertainty.[95] In lieu of first-hand disclosures of inference's environmental footprint, researchers are forced to make a series of difficult discretionary choices with major implications for their findings.[96] In addition, no standardized way to report findings has emerged; some researchers report that they index findings "by query" while others use specific token counts or GPU hours.[97] The range of variables that meaningfully affect estimates makes it nearly impossible to generate direct comparisons. Given the underlying challenges in assessment, this section assesses the research landscape, rather than identifying a single most accurate assessment, suggesting that academics as well as the public are largely unable to determine the true costs of AI inference.[98]

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95. *See* Ian Schneider et al., *Life-Cycle Emissions of AI Hardware: A Cradle-To-Grave Approach and Generational Trends*, ARXIV (Feb. 2025), https://arxiv.org/pdf/2502.01671 (modeling how to assess the lifecycle impacts of AI); Lynn H. Kaack et al., *Aligning Artificial Intelligence with Climate Change Mitigation*, 12 NATURE CLIMATE CHANGE 518 (2022) (offering a "systematic framework for describing the effects of machine learning (ML) on GHG emissions"); O'Donnell & Crownhart, *supra* note 5 (explaining how estimating AI's environmental footprint is nearly impossible without more transparency from AI companies and data center operators). Most research on AI inference measures compute costs from running the programs but does not account for cooling and other data center operations that can make up as much as 50% of energy demand associated with a given inference. Research on the energy costs associated with AI often distinguishes between the energy used to run the programs and the energy used in data center operations like cooling. This paper uses the doubling approach to account for non-compute energy consumption and refers to these figures as "secondary costs." *See* O'Donnell & Crownhart, *supra* note 5 (citing Pratyush Patel et al., *Characterizing Power Management Opportunities for LLM's in the Cloud*, MICROSOFT AZURE (Apr. 27, 2024), https://www.microsoft.com/en-us/research/wp-content/uploads/2024/03/GPU_Power_ASPLOS_24.pdf ("doubling the amount of energy used by the GPU gives an approximate estimate of the entire operation's energy demands").

96. For example, whether a researcher assumes that an inference is being powered by a Nvidia A100, H100, or B200 GPU can make the resulting energy estimate vary by more than an order of magnitude. *See What Is MLPerf?*, NVIDIA CORP., https://www.nvidia.com/en-au/data-center/resources/mlperf-benchmarks/ (last visited Aug. 11, 2025). The problem is, unless researchers are running the model on their own hardware (which has its own tradeoffs in terms of accuracy) they cannot possibly know what kind of chip(s) processed their query.

97. GPU hours is measured by multiplying the number of GPUs running by the hours they are running (i.e., 1 GPU hour = 1 GPU running for 1 hour, 2 GPUs running for 30 minutes, etc.). The energy consumption of one GPU hour is variable.

98. To control for at least one of these variables, this section pulls research conducted since November 2022. This avoids irrelevant findings based on outdated model constructions. Although any discrete cutoff is somewhat arbitrary, the release of GPT-3 in November 2022 marked a significant change in the AI landscape.

The most commonly cited estimates of AI energy demand are 3Wh and 2.9Wh per query.[99] The problem is that these numbers are completely contrived.[100] Per Dr. Sasha Luccioni, the first estimate dates to a 2009 blog post from Google[101] and a 2023 interview with Alphabet Chairman John Hennesy, who has no affiliation with OpenAI.[102] A 2023 SemiAnalysis report seeking to estimate the financial cost of integrating ChatGPT-like functionality into every Google search spawned another figure when researcher Alex de Vries extrapolated the report's findings to estimate energy consumption.[103] De Vries himself later cast doubt on the accuracy of his calculation, saying it "felt like 'grasping at straws' . . . because he had to rely on third-party estimates he could not replicate."[104] Notably, the original SemiAnalysis article does not reference energy consumption. The popularity of these figures despite their weaknesses indicates the demand for clear data about AI energy use and the opaque information ecosystem confronting researchers.

Even if these estimates were accurate at the time, they have quickly become outdated due to hardware and software developments. The Nvidia A100 chip, used by SemiAnalysis in the 2023 report, has been succeeded by two new generations of chips (known as Hopper and Blackwell chips) that boast significantly more computational power per watt.[105] Further, Google's TPUs (Tensor Processing Units) and Groq's LPUs (Language Processing Units) are optimized for the math and algorithms that power LLMs (both chips are specifically optimized to process the complex linear algebra equations that power LLMs whereas GPUs were originally designed for 3D graphics that require slightly different equations) and claim to be even more efficient than Nvidia GPUs.[106]

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99. Luccioni et al., *supra* note 57, at 4–5 (finding that 53% of news reports on "ChatGPT energy consumption" cite the 3Wh figure or claim it consumes 10 times more energy than a Google search); Alex de Vries, *The Growing Energy Footprint of Artificial Intelligence*, 7 JOULE 2191 (Oct. 18, 2023) (estimating 2.9 Wh per request). De Vries's paper has been cited hundreds of times.

100. *See* Luccioni et al., *supra* note 57, at 4–5 (laying out several key uncertainties in the estimate).

101. *Id.* (citing Urs Hölzle, *Powering a Google Search*, GOOGLE: OFFICIAL BLOG (Jan. 11, 2009), https://googleblog.blogspot.com/2009/01/powering-google-search.html).

102. *Id.* (citing Jeffrey Dastin & Stephen Nellis, *Focus: For Tech Giants, AI Like Bing and Bard Poses Billion-Dollar Search Problem*, REUTERS (Feb. 22, 2023), https://www.reuters.com/technology/tech-giants-ai-like-bing-bard-poses-billion-dollar-search-problem-2023-02-22/).

103. de Vries, *supra* note 99 (citing Dylan Patel & Afzal Ahmad, *The Inference Cost Of Search Disruption – Large Language Model Cost Analysis*, SEMIANALYSIS (Feb. 9, 2023), https://semianalysis.com/2023/02/09/the-inference-cost-of-search-disruption/).

104. Sophia Chen, *AI's Energy Problem*, 639 NATURE 22, 23 (2025).

105. *See NVIDIA Blackwell Platform Arrives to Power a New Era of Computing*, NVIDIA CORP. (Mar. 18, 2024), https://nvidianews.nvidia.com/news/nvidia-blackwell-platform-arrives-to-power-a-new-era-of-computing (detailing the capabilities of NVIDIA's new Blackwell chip, which succeeded the Hopper architecture).

106. Molly McHugh-Johnson, *Ask a Techspert: What's the Difference Between a CPU, GPU, and TPU?*, THE KEYWORD (Oct. 30, 2024), https://blog.google/technology/ai/difference-cpu-gpu-tpu-trillium/; Andrew Ling, *Inside the LPU: Deconstructing Groq's Speed*, GROQ (Aug. 1, 2025), https://groq.com/blog/what-nvidia-didnt-say; Kurtis Pykes, *Understanding TPUs vs GPUs in AI: A Comprehensive Guide*, DATACAMP (May 30, 2024), https://www.datacamp.com/blog/tpu-vs-gpu-ai ("In contrast to

As software grows more complex, demanding more functions to fulfill a given request, more efficient chips will not necessarily decrease total energy consumption. Rather, the energy-efficiency of modern chip hardware will be offset by the greater energy demand of ever-expanding models. For instance, a team led by Sasha Luccioni in 2024 observed a relationship between the size of a model and energy consumption due to the additional computational intensity of larger models.[107] The largest model of two different architectures studied by the Luccioni team (Flan-T5-xxl and BLOOMz-7B) consumed 3.2 and 1.9 times as much energy, respectively, as the smallest models of the same architecture (Flan-T5-base and BLOOMz-560M) across all task types.[108] Research by teams led by Siddharth Samsi in 2023 and Erik J. Husom in 2024 corroborated the positive relationship between model size and energy consumption.[109] Further evidence that model size may counteract chip efficiency improvements comes from a 2025 estimate that OpenAI's "largest and best" model, GPT-4.5,[110] consumes 6.72Wh per request, despite the authors' assumption that GPT-4.5 loads are processed on H100 or H200 chips, which are more efficient than the A100s used in previous research.[111]

Other design choices, such as whether a model is locally operated, can complicate the relationship between model size and energy consumption. As the authors in the GPT-4.5 study point out, its high energy consumption "suggests inefficiencies rooted in model architecture."[112] In the first study of OpenAI's open source GPT-OSS models co-created with Hugging Face, researchers found that the

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GPUs, which were initially created for graphics processing tasks and subsequently modified to cater to the demands of AI, TPUs were designed specifically to accelerate machine learning workloads.").

107. Alexandra Sasha Luccioni, Yacine Jernite & Emma Strubell, *Power Hungry Processing: Watts Driving the Cost of AI Deployment?*, ARXIV (Oct. 15, 2024), https://arxiv.org/pdf/2311.16863. To clarify, they draw a relationship between model size and emissions, but because they calculate emissions by multiplying energy consumption by a constant emissions factor, the same relationship would exist for energy. The study found that task structure (i.e., the kind of task the LLM is asked to perform) is more consequential than model size, although both were found to have an influence. We discuss this in more detail in the subsequent section.

108. *See* Luccioni, Jernite & Strubell, *supra* note 107, at 10 (Flan-T5-xxl consumed 0.083kWh per 1,000 responses compared to 0.026kWh for Flan-T5-base. BLOOMz-7B consumed 0.104kWh per 1,000 responses compared to 0.054kWh per 1,000 responses for BLOOMz-560M).

109. *See* Siddhartha Samsi et al., *From Words to Watts: Benchmarking the Energy Costs of Large Language Model Inference*, ARXIV (Oct. 4, 2023), https://arxiv.org/pdf/2310.03003 (finding that larger LLaMa models consistently produce less words per second and consume more energy per second for multiple tasks, datasets, and types of chips); Erik Johannes Husom et al., *The Price of Prompting: Profiling Energy Use in Large Language Models Inference*, ARXIV (Jul. 4, 2024), https://www.arxiv.org/pdf/2407.16893 (finding the same results across various models).

110. *Introducing GPT-4.5*, OPENAI (Feb. 27, 2025), https://openai.com/index/introducing-gpt-4-5/.

111. Jegham, *supra* note 81. Importantly, the query length (100 token prompt, 300 token response) is much longer than Luccioni et al. 2024 (54 token prompt, 8.52-10.68 token response). Though it may play a role, it is very unlikely that the difference in prompt and response length accounts for most or all of the discrepancy.

112. *Id.* at 7.

20B parameter model consumed 2.02Wh per 100-token response.[113] Meanwhile, the 120B model consumed 8.31Wh per 100 tokens.[114] The fact that these newer models were estimated to consume much more energy than previous models while running on A100s suggests that complexity is increasing and could either offset hardware efficiency gains or, more likely, result in net increases in energy consumption until data centers upgrade to newer models. The 20B model is particularly interesting because it was designed to run on everyday consumer devices without a dedicated GPU.[115] It is probable that running models locally will increase inference energy consumption because individuals' devices are not optimized in the same way as data centers.[116] Furthermore, if models migrate to operate locally, it could further confuse efforts to trace AI energy consumption, since consumption will be spread across millions of users who are unlikely to track their computer's energy.

OpenAI's decision to automatically route prompts to different models based on private, proprietary criteria further exacerbates the limitations of per-query estimates – especially if the practice becomes common. This routing algorithm does not allow the user to choose the inference model, nor to find out which model will be used. Thus, the uncertainty in calculating emissions per query compounds both the variation within each inference model from one query to another and the variation in which model a query will be routed to.[117]

B. *The Future of AI Energy Demand: Developments and Likely Directions*

Several new developments suggest that other factors may be even more consequential. First, the introduction of reasoning models may substantially increase energy demand.[118] Unlike traditional models that predict the next token based on probability, reasoning models introduce a "thinking" phase in which they break the problem into several steps and analyze responses along the way.[119] The additional thinking stage included in these models means they perform additional computations before arriving at a response. Three of the four models with the highest energy consumption studied by a team led by Nidhal Jegham in 2025 are reasoning models,

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113. Sasha Luccioni, *The GPT-OSS Models are Here . . . and They're Energy-Efficient!*, HUGGING FACE BLOG (Aug. 7, 2025), https://huggingface.co/blog/sasha/gpt-oss-energy.

114. *Id.*

115. OPENAI, *supra* note 60.

116. The efficiency of consumer devices can vary greatly, but research comparing the energy demand of cloud computing to on-site computing finds significant savings from moving to data centers. *See* Jiyong Park, Kunsoo Han & Byungtae Lee, *Green Cloud? An Empirical Analysis of Cloud Computing and Energy Efficiency*, 69 MGMT. SCI. 1639 (2023); *see also* Masley, *supra* note 58; discussion *supra* note 68.

117. *See* O'Donnell & Crownhart, *supra* note 5 (noting the uncertainty around what model is used to answer a query adds to the complexity of calculating emissions).

118. *See* Vinayakh, *supra* note 46 (describing how reasoning models work and how they differ from traditional LLMs).

119. *Id.*

with Deepseek R1 leading the pack at 23.82Wh per response.[120] Although reasoning is a powerful tool for some applications that require advanced problem solving, it is unnecessary for many common functions (e.g., seeking specific information or personal writing), so defaulting to these more powerful models may increase AI's energy demand without meaningfully improving the output.[121]

Second, and most important, is the evolution in the tasks that AI tools are being asked to complete. Text-to-text functions (like asking a question to a chatbot) tend to be orders of magnitude less consumptive than image or video generation. Image generation consumed nearly 60 times as much energy per response than the most intensive text-to-text operation studied by the Luccioni team (5.81Wh and 0.098Wh, respectively).[122] More recently, researchers estimated that Stable Diffusion 3 Medium consumes 1.22Wh per image, significantly less than the Luccioni team's original calculation but still far more consumptive than text-to-text functions on similarly sized models.[123] Whereas text-to-text tasks usually have the model generate a string of text based on the probability that one token will follow its predecessor, image and video generation usually involves thousands of inferences over dozens of "inference" or "diffusion steps." These "diffusion models" begin with random noise before gradually refining the image over many iterations (i.e., diffusion steps).[124] More diffusion steps typically lead to a superior output, but they also tend to be more energy intensive since the model is effectively performing additional individual inferences before returning an image.[125]

Video generation is likely even more intensive, but research about its specific demand is sparse. In 2025, a team led by Julien Delavande studied open-source text-to-video models and found that energy consumption for a single response ranged by almost three orders of magnitude, from 0.28-218.94Wh.[126] The researchers

120. Jegham, *supra* note 81, at 8. The other reasoning models, OpenAI o3 and o1, consumed 7.03 and 4.45Wh/response, respectively. *Id.* GPT-4.5 was the third most consumptive despite note being a reasoning model due to its large size and "inefficiencies rooted in model architecture." *Id.* at 7.

121. *See Reasoning Best Practices*, OPENAI PLATFORM, https://platform.openai.com/docs/guides/reasoning-best-practices (last visited Apr. 11, 2026) (explaining when it may be best to use more intense reasoning models versus the "workhorse" non-reasoning models).

122. Luccioni, Jernite & Strubell, *supra* note 107, at 6 (figures doubled to account for secondary costs).

123. Based on the cost to generate a standard-quality 1024x1024 pixel image using 50 diffusion steps with a model that has 2B parameters. O'Donnell & Crownhart, *supra* note 5.

124. *See* Dave Bergman & Cole Stryker, *What are Diffusion Models?*, IBM (Aug. 21, 2024), https://www.ibm.com/think/topics/diffusion-models (explaining more about diffusion models).

125. O'Donnell & Crownhart, *supra* note 5 ("[D]oubling the number [of] diffusion steps to 50 just about doubles the energy required.") Output quality probably experiences decreasing marginal returns relative to energy consumption, implying that additional diffusion steps are not always necessary. *See* Zeyu Yang, Karel Adámek & Wesley Armour, *Double-Exponential Increases in Inference Energy: The Cost of the Race for Accuracy*, 10 ARXIV (Dec. 12, 2024), https://arxiv.org/html/2412.09731v1 ("[W]hile state-of-the-art models achieve higher accuracy, they incur significant energy costs with diminishing returns.").

126. Julien Delavande, Sasha Luccioni & Régis Pierrard, *How Much Power does a SOTA Open Video Model Use?*, HUGGING FACE BLOG (Jul. 2, 2025), https://huggingface.co/blog/jdelavande/text-to-video-energy-cost (figures doubled to account for secondary costs).

explained that factors like model size, the number of diffusion steps, resolution, video length, and architectural design are responsible for the discrepancy, but it is probable that some level of uncertainty also contributed to the results' wide range.[127] Increasing these factors can improve the video that the model generates by making it longer, higher resolution, and more responsive to the original prompt, but these improvements come with a high cost.[128] Given the dazzling ability of the newest closed-source models like OpenAI's Sora and Google's Veo 3, it is almost certain that generating videos using these models is more energy intensive than even the highest estimate included in the 2025 article by the Delavande research team.[129]

We were unable to find any studies directly comparing the inference costs of traditional LLMs to the inference costs of multi-modal large language models (MLLMs). An MLLM can perform multiple task types, including text, image, and video.[130] It is reasonable to assume that some relationship exists between the number of task structures an MLLM can perform and its energy consumption for a given task, because MLLMs typically require more parameters and computational intensity. Future research should focus on comparing the energy demand for multi-modal and unimodal models when performing the same task.

Finally, agentic AI deployment could drastically increase total energy demand for AI. An agent, as opposed to a traditional or reasoning model, combines dynamic reasoning with the use of external tools to independently perform a specific function with minimal human interaction.[131] For example, a day trader might create an AI agent to automatically execute transactions based on live market data, or a company might automate customer service by giving a chat bot access to a customer's

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127. *Id.*

128. Whereas Delavande, Luccioni & Pierrard, *supra* note 126 estimated that a 6.13-second video with CogVideoX used 51.86Wh (after accounting for non-compute costs), the creators of the model reported that generating a shorter (5-second) video with higher resolution required a H800 chip to run for 500 seconds (an Nvidia chip designed to comply with export controls to China). The H800 has a maximum power draw of 700W. Over a 500-second period, this means generating the shorter but more refined video used up to 97.2Wh. The discrepancy may be in part due to the hardware available to each team, but it also points to the additional energy costs of higher resolution video generation. *See* Zhuoyi Yang et al., *CogVideoX: Text-to-Video Diffusion Models with an Expert Transformer*, ARXIV at 16 (Mar. 26, 2025), https://arxiv.org/pdf/2408.06072.

129. *See* Yang et al., *Double-Exponential Increases in Inference Energy: The Cost of the Race for Accuracy*, 10 ARXIV (Dec. 12, 2024), https://arxiv.org/html/2412.09731v1 ("[W]hile state-of-the-art models achieve higher accuracy, they incur significant energy costs with diminishing returns."); Julien Delavande, Regis Pierrard & Sasha Luccioni, *Video Killed the Energy Budget: Characterizing the Latency and Power Regimes of Open Text-to-Video Models*, ARXIV (Sep. 23, 2025), https://arxiv.org/pdf/2509.19222 (finding that energy consumption for video generation scales geometrically with spatial resolution and number of frames, and linearly with the number of diffusion steps). All of these variables are means to improve the quality of video.

130. Jobit Varughese, *What is a Multi-Modal LLM (MLLM)?*, IBM, https://www.ibm.com/think/topics/multimodal-llm (last visited Apr. 11, 2026).

131. Kiin Kim et al., *The Cost of Dynamic Reasoning: Demystifying AI Agents and Test-Time Scaling from an AI Infrastructure Perspective*, ARXIV at 2 (Jun. 4, 2025), https://arxiv.org/pdf/2506.04301 (defining AI agents).

profile.[132] Potential applications of agents are broad, with some agents expected to run continuously as they perform a given task. Since agents perform so many calculations to complete a task, they may be many times more energy consumptive than non-agentic models.[133] One study found that agent-augmented LLaMa-3.1-Instruct 70B and LATS used between 62.1 and 136.5 times as much energy to complete the same task as the base model, respectively (2.55Wh per query compared to 158.48 and 348.41Wh per query).[134] The technology is very new, meaning that only limited research has focused on its energy consumption, and efficiency improvements are likely. That said, if agents rapidly overtake traditional models, we may see massive growth in AI energy load.

The addition of more steps (reasoning, tool use, diffusion steps, etc.) is one universal factor of increased energy intensity among AI developments. These steps improve response quality in many instances, but they also add computational load. Reasoning models outperform traditional models on many tasks, and more diffusion steps improve image quality.[135] It is unclear, though, whether these improvements are sufficient to justify the cost for every task. Further, their widespread deployment may unnecessarily drive up the energy demand and cost of AI inference. Reasoning is unnecessary for many simple tasks, and the number of diffusion steps reaches a point where improvements are unnoticeable. For example, AI writing assistants do not need reasoning for spellcheck or grammar suggestions and generating an AI avatar does not need 150 diffusion steps to be acceptable. Greater transparency about energy use may induce developers to critically assess the deployment of their models before defaulting to higher energy designs.[136]

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132. *See generally* Matt Renner & Matt A.V. Chaban, *1001 Real-World Gen AI Use Cases From the World's Leading Organizations*, GOOGLE BLOG (Oct. 9, 2025), https://cloud.google.com/transform/101-real-world-generative-ai-use-cases-from-industry-leaders (discussing 1001 use cases of AI agents).

133. Clive Thompson, *Coding after Coders: The End of Computer Programming as We Know It*, N.Y. TIMES (Mar. 12, 2026) https://www.nytimes.com/2026/03/12/magazine/ai-coding-programming-jobs-claude-chatgpt.html (reporting that programmers use AI agents to execute programming tasks, with one agent supervising other agents that write code and still other agents that test the code, so one query can invoke large numbers of agents to perform tasks, with each agent adding tasks depending on what other agents do). Moreover, the ease of delegating programming tasks creates something like a Jevons paradox that further increases the amount of computing AI agents are asked to perform: "Because the agents can produce functioning code so quickly, their human overseers can experiment, trying things out to see what works and discarding what doesn't. . . .The fact that A.I.'s output can be unpredictable — if you ask it to write a piece of code, it might do so in a slightly different way each time — 'is addictive, in a slot-machine way.'" *Id.*

134. Kim et al., *supra* note 131, at 10.

135. *See* Zhong-Zhi Li et al., *From System 1 to System 2: A Survey of Reasoning Large Language Models*, ARXIV (Feb. 24, 2025), https://arxiv.org/html/2502.17419v1 (on reasoning); Yongshi Ye et al., *How Well Do Large Reasoning Models Translate? A Comprehensive Evaluation for Multi-Domain Machine Translation*, ARXIV (May 26, 2025), https://arxiv.org/html/2505.19987v1 (on reasoning); Chen Hou, Guoqiang Wei & Zhibo Chen, *High-Fidelity Diffusion-based Image Editing*, ARXIV (Jan. 4, 2024), https://arxiv.org/html/2312.15707v3 (on image generation).

136. Similar efforts have proven effective in physical designs. *See, e.g.*, Tripp Shealy et al., *Bringing Choice Architecture to Architecture and Engineering Decisions: How the Redesign of Rating Systems Can Improve*

C. AI Footprint Calculators

AI footprint calculators offer a valuable approach to disclosure, as they are designed to be accessible to a broad audience.[137] By comparing multiple models, footprint calculators can serve as a one-stop resource for those interested in understanding the energy and environmental costs of AI inference. It is easy to assume that footprint calculators are just for retail users of AI, but they have the potential to educate companies, governments, the managers responsible for AI sourcing, household users, policymakers, and even AI firms themselves. The accuracy of these footprint calculators is essential, though, as footprint calculators that are flawed or based on incorrect assumptions can lead users to make poor decisions about the best AI firms, models, or queries for their needs. This section examines four footprint calculators to explore the differences in their estimates of AI inference's energy consumption, the implications of these discrepancies, and why they may occur.[138]

Sustainability, 35 J. MGMT. ENG'G (2019) (suggesting that building sustainability rankings into the interfaces designers use is a particularly effective way to induce sustainability improvements in the end product).

137. Calculators have been used in many contexts to achieve similar goals. *See, e.g.,* E. Röös et al., *Introducing a Comprehensive and Configurable Tool for Calculating Environmental and Social Footprints for Use in Dietary Assessments*, 519 J. CLEANER PROD. 146002 (2025) (providing a calculator to assess the environmental impacts of foods and diets); J. Paul Padgett et al., *A Comparison of Carbon Calculators*, 28 ENV'T IMPACT ASSESSMENT REV. 106 (2008) (evaluating carbon footprint calculators).

138. This section does not cover LLM Perf Leaderboard which is popular but has limited model coverage. *See LLM-Perf Leaderboard*, HUGGING FACE BLOG, https://huggingface.co/spaces/optimum/llm-perf-leaderboard (last visited Aug. 12, 2025).

Table 1: Characteristics of selected AI footprint calculators.[139]

Calculator	Ecologits[140]	AI Energy Score[141]	ML.ENERGY[142]	ChatUI[143]
Disclosure Type	Individual Select	Multi compare	Multi compare	Live tracker
Disclosure Units	Energy (Wh) to complete a task (e.g. write a 50-token Tweet)	GPU energy (Wh) per 1,000 responses	Energy (J) per response	Total energy (Wh) and duration (sec)
Function Type	Wh/Token	Wh/response	Wh/response	Wh/second[144]
Scope	Compute and Cooling	GPU energy	GPU energy	GPU energy
Source Database(s)	N/A[145]	WikiText, OSCAR, UltraChat[146]	ShareGPT[147]	N/A
Task Coverage	Text	Multimodal	Multimodal	Text
Hardware	NVIDIA A100 80GB[148]	NVIDIA RTX 4090 (<20B) NVIDIA H100 (>20B)[149]	NVIDIA H100 80GB[150]	NVML API, average power ≈70W

Differing design choices in footprint calculators make direct comparison challenging (see Table 1 for an overview of key characteristics of each footprint

139. It is impossible to completely control for response length (i.e. how many tokens make up one "response") and each footprint calculator is slightly different in this regard.

140. CODECARBON, *supra* note 36.

141. Sasha Luccioni & Boris Gamazaychikov, *AI Energy Score Leaderboard*, HUGGING FACE BLOG (Dec. 2025) https://huggingface.co/spaces/AIEnergyScore/Leaderboard.

142. ML.ENERGY LEADERBOARD, *supra* note 39.

143. Delavande, *supra* note 36.

144. Qwen3 8B uses real energy consumption from NVIDIA NVML but all others are a function of estimated average power and response time. Since total energy is a linear time function, we can convert from seconds to tokens/responses by tokenizing outputs then dividing total energy by number of tokens.

145. Energy is benchmarked using LLM-Perf Leaderboard based on model architecture and number of output tokens CODECARBON, *supra* note 36.

146. Luccioni & Gamazaychikov, *supra* note 141. Total input tokens (T) for text generation = 369,139 for 1,000 data points. *Id.* We could not identify the average tokens per response, but, based on the study design, we assume it is around 369T.

147. Average output tokens specified by model when selecting "Advanced columns," range between 351-1464 tokens for all models. ML.ENERGY LEADERBOARD, *supra* note 39.

148. EcoLogits released an update in November 2025 that updated models to H100 GPUs. As mentioned previously, the analysis in this paper is based on the original work done in August 2025. The change reflects the rapid movement in AI hardware and software development. *See Environmental Impact of LLM Interference,* ECOLOGITS (2025), https://ecologits.ai/0.8/methodology/llm_inference/#on-hardware.

149. Luccioni & Gamazaychikov, *supra* note 141.

150. In a December 2025 update, ML.Energy Leaderboard added B200 as a hardware option. At the time of initial analysis, the H100 was the most recent hardware available. Version history is available in the "About" box at ML.ENERGY LEADERBOARD, *supra* note 39.

calculator). This Article opts for a per-token approach to standardize estimates by conforming data to energy consumption per 400-term response (Wh/400T). It also compares similar models since all four footprint calculators only share one model in common (LLaMa 3.1 8B),[151] although they may cover different generations of the same model (e.g., LLaMa 3.1 70B and LLaMa 3.3 70B). It is important to compare several models to identify how methodological differences scale across LLMS. Comparisons, therefore, will not be perfect but will identify major differences in the estimates.

Differences between the footprint calculators are immediately clear from the LLaMa 3.1 8B estimates (see Table 2 below). The highest estimate was 58 times greater than the lowest, and both multi-compare calculators estimated that the models consumed at least 30 times less energy than Ecologits or ChatUI.[152]

The discrepancies are not only about the magnitude of emissions, but also reflect the calculators' varying estimates for each model's energy efficiency. EcoLogits estimates that LLaMa 3.1 8B consumes 58% as much energy as Gemma 2/3 27B and LLaMa 3.3 70B consumes more than 3 times the energy of LLaMa 3.1 8B, but ChatUI estimates the opposite: that LLaMa 3.1 8B consumes more than twice as much energy as both Gemma 2/3 27B, and LLaMa 3.3 70B. When converted to a standard 400-token response, ChatUI Energy provides estimates that are most consistent with studies of similarly sized models from the same generation, but these estimates are still several times higher than estimates from the literature.[153]

The relative difference between the highest and lowest estimates for the larger LLaMa 3.1 70B model is smaller, but the gross difference is much larger.[154] A person who generates 1,000 tokens of text each day for a year with LLaMa 3.1 70B could believe that they are using anywhere from 0.14 to 4.6 kWh, depending on the footprint calculator they reference. For the even larger 3.1 405B model, the same person could believe that they are using anywhere from 1.5 to 87 kWh per year.[155] When scaled to the grid or interconnection level, these discrepancies can lead decision-makers to reach vastly different conclusions about which AI investments might be warranted. Model providers can help those responsible for managing energy supply to make informed choices by disclosing their marginal energy demand.

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151. ChatUI Energy uses Nous Research's fine-tuned version of LLaMa models. *See* Ryan Teknium, Jeffrey Quesnelle & Chen Guang, *Hermes 3 Technical Report*, ARXIV (Aug. 15, 2024), https://arxiv.org/pdf/2408.11857 (explaining the fine-tuned versions of the LLaMa models).

152. Ecologits = 3.78E-3Wh/T, ChatUI = 1.98E-3Wh/T, AI Energy Score = 9.42E-5Wh/T, ML.ENERGY = 6.5E-5Wh/T. All figures except Ecologits doubled to account for secondary costs.

153. Estimate = 0.79Wh/request. *See, e.g.*, Wong, *supra* note 85; Husom et al., *supra* note 109.

154. ML.ENERGY Leaderboard had the lowest again at 3.77E-4Wh/T and Ecologits had the highest at 1.26E- 2Wh/T.

155. 0.239Wh/T for Ecologits, 4.14E-3 for ML.ENERGY Leaderboard. Neither of the other footprint calculators cover 3.1 405B.

Table 2: Comparison of estimates of selected models in Wh/400T:
**Indicates figures have been doubled to account for secondary costs.*[156]

	EcoLogits	AI Energy Score*	ML.ENERGY*	ChatUI*
Phi 3 Mini	1.23	N/A	0.0257	N/A[157]
LLaMa 3.1 8B	1.51	0.0376	0.0260	0.79
Gemma 2/3 27B	2.57	N/A	0.0709	0.32
Mistral 8x7Bv0.1[158]	1.73	1.33	0.0945	N/A
LLaMa 3.1/3.3 70B	5.02	3.73	0.15	0.32

Comparing ChatUI's estimates for LLaMa 3.1 8B and LLaMa 3.3 70B shows how design choices can influence the estimates generated by a footprint calculator. ChatUI's equation using average power draw and duration make the estimated energy per token for LLaMa 3.3 70B less than LLaMa 3.1 8B because of LLaMa 3.3's superior token throughput.[159] In part, this reflects Meta's intention to make LLaMa 3.3 a more efficient option that can operate with fewer calculations (and therefore produce tokens faster),[160] but the estimate relies on the improbable assumption that both models have the same average power draw, undercutting the reliability of the estimate.

Ecologits' reliance on model size, meanwhile, may overlook key factors like computational intensity.[161] EcoLogits estimates GPU energy by averaging the number of active parameters, GPUs needed to service a model based on its active parameters, average output tokens based on model size, and tokens per second (i.e., generation latency) of a variety of models. EcoLogits then adds this result to an estimate of server energy *without* GPUs and multiplies the figure by a power usage effectiveness (PUE) of 1.2.[162] The assumptions that different models of all different

156. See footnote 98 for an explanation of secondary costs and why they are necessary to consider.

157. Phi 3.5 Mini is listed but when prompted returns an error saying the model does not exist.

158. Assumes that not all parameters are active. It is possible to change this in "Expert Mode," but the default is presented here.

159. 8.06E-4Wh/T for 3.3. Both models are assumed to draw 1.53E-2Wh/sec, but 3.3 can produce more than twice as many tokens per second.

160. Ahmad Al-Dahle (@Ahmad_Al_Dahle), X (Dec. 6, 2024, 11:30 AM), https://x.com/Ahmad_Al_Dahle/status/1865071436630778109 (noting that Llama 3.3 70B is designed to be "easier & more cost-efficient to run").

161. For instance, it is unlikely that LLaMa 2 7B and Mistral 7B v0.3 consume the same energy per token but Ecologits assumes as much.

162. ECOLOGITS, *supra* note 148; PUE is a metric to assess computational efficiency. It is the ratio between total energy consumption and energy used for computation. Higher figures are less efficient (i.e., a data center with PUE 3.0 would consume 2Wh in overhead for every 1Wh for computation) and 1.0 is the theoretical ideal (i.e., every bit of energy is used for computation). The average data center in the US

sizes are similar enough to accurately predict a trend and that every model with the same number of active parameters will consume the same amount of energy to fulfill a task are likely untrue. With that said, these assumptions, accurate or not, allow EcoLogits to predict the energy demand of the most popular models like ChatGPT since researchers cannot run proprietary models locally.

AI Energy Score and ML.ENERGY Leaderboard may misrepresent actual usage patterns since they gather estimates from a random assortment of queries. A 2025 study by a team led by Marta Adamska found that the "semantic meaning of the prompt" plays a role in the associated energy consumption.[163] For instance, prompts that include the keyword "analyze" were found to noticeably increase energy consumption relative to prompts that used the keyword "classify" because they prime the model to respond differently.[164] If the dataset does not reflect the ways individuals tend to use AI tools, including the words and their implied meanings, it may cause the estimates to err.[165]

In sum, footprint calculators are important tools, but they suffer from data availability problems and from the inability to validate their estimates by comparing them to actual measurements of emissions or energy consumption.[166] They produce results that differ from each other based on the same inputs and from the conclusions in the academic literature because they must make assumptions without a standard method for estimating AI energy use. Ideally, disclosure from model providers would make footprint calculators obsolete, but in the meantime a viable alternative is to develop a standardized calculation methodology that accounts for as many relevant factors as possible. A standardized methodology will help avoid inaccurate or misleading information in these public-facing tools.

III. AS COMPARED TO WHAT?

Proponents of AI proliferation often compare its energy use to that of its alternatives. For example, if one AI prompt can get a user to the same place as 30

has a PUE of 1.4, though PUE can vary widely between locations and over time. Shehabi et al., *supra* note 8, at 39–40, 47.

163. Marta Adamska et al., *Green Prompting*, ARXIV, 7 (Mar. 9, 2025), https://arxiv.org/html/2503.10666v1.

164. *Id.* at 6.

165. ML.ENERGY Leaderboard used real user prompts to try to control for usage patterns, but the dataset likely suffers from selection bias and has since gone out of service, so it is impossible to know how well their dataset predicts real-world patterns. *See* SHAREGPT, www.sharegpt.com (last visited Aug. 12, 2024).

166. It is important to note that AI companies themselves lack a standardized methodology to accurately assess their AI energy consumption. *See* O'Donnell & Crownhart, *supra* note 5. These challenges compound the difficulties for researchers. Preliminary efforts like those from Mistral and Google provide important benchmarks nonetheless. *See* MISTRAL AI, *supra* note 91; Elsworth et al., *supra* note 41.

Google searches, it is likely that AI is the more efficient option.[167] Likewise, if AI-assisted coding can produce the same code in a fraction of the time that a programmer can, it may yield net energy savings by reducing the amount of time a programmer's screen is on[168] or the hours spent in an air-conditioned office. AI inference is relatively efficient by all accounts, so it may be able to complete tasks using far less energy than its alternatives.

Applying AI to optimize inefficient systems and accelerate research into low-energy solutions may also yield significant energy savings.[169] Companies like Google and NextEra have developed AI tools to eliminate unnecessary consumption and boost clean energy generation.[170] Researchers at the Paul Sherrer Institute, meanwhile, have used AI to rapidly conceive of and evaluate thousands of formulations for cleaner concrete recipes.[171] Using AI thus may not just involve immediate tradeoffs but also the potential to accelerate progress in other areas. It is possible that seemingly unrelated developments today will reveal new solutions to difficult climate problems, just as semiconductors have become an essential part of solar energy despite originally serving a very different purpose.[172]

Although it is true that AI is often efficient, especially when considering its complexity, the cumulative energy consumption of billions of inferences is already

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167. *See* Matthew Elmore, *The Hidden Costs of ChatGPT: A Call for Greater Transparency*, 23 AM. J. BIOETHICS 47, 48 (2023) (concluding that a single AI prompt has the carbon footprint of five Google searches). This suggests that AI could be six times more efficient.

168. Computer monitors typically up to 26W, so reducing the screen time of a monitor by just a few minutes may balance out the energy from an AI query. *See, e.g.*, *ENERGY STAR Certified Displays*, ENERGY STAR, https://www.energystar.gov/productfinder/product/certified-displays/ (last visited Apr. 11, 2026).

169. *See, e.g.*, Stern et al., *supra* note 74; *see also* Daniel Slate et al., *Adoption of Artificial Intelligence by Electric Utilities*, 45.1 SSRN J. 1 (2024) (discussing the role of AI adoption in optimizing electric utilities); Stein, *supra* note 5, at 902 ("Google and DeepMind applied machine learning to wind power capacity in the United States 'to better predict wind power output thirty-six hours ahead of actual generation,' assisting in optimal hourly day-ahead delivery commitments.").

170. Google's Alpha Evolve does this by automatically improving complex algorithms and has helped Google recover 0.7% of its global compute. *AlphaEvolve: A Gemini-Powered Coding Agent for Designing Advanced Algorithms*, GOOGLE DEEPMIND (May 14, 2025), https://deepmind.google/discover/blog/alphaevolve-a-gemini-powered-coding-agent-for-designing-advanced-algorithms/. Google DeepMind's prediction software helped Google better predict wind conditions and thereby increase the value of its wind generation. *See* Carl Elkin & Sims Witherspoon, *Machine Learning can Boost the Value of Wind Energy*, GOOGLE DEEPMIND (Feb. 26, 2019), https://deepmind.google/blog/machine-learning-can-boost-the-value-of-wind-energy/. NextEra claims that it is leveraging AI to help customers minimize the deployment of backup fossil fleets. *See Meeting Unprecedented Power Demand*, NEXTERA RESOURCES, https://www.nexteraenergyresources.com/resource-center/insights/electricity-demand-is-on-the-rise.html (last visited Feb. 14, 2026).

171. Romana Boiger et al., *Machine Learning-accelerated Discovery of Green Cement Recipes*, 58 SPRINGER NATURE 173 (Apr. 25, 2025).

172. *See* Martin Green, *Silicon Photovoltaic Modules: A Brief History of the First 50 Years*, 13 PROGRESSIVE PHOTOVOLTAICS: RSCH. & APPLICATIONS, 447, 447–48 (2005) (noting that semiconductors were initially developed to power telecommunications systems); *see also* Stern et al., *supra* note 74 (explaining how AI has led to insights about protein folding and crystal construction that have major implications for alternative protein sources and renewable energy, respectively).

significant and expected to increase. The marginal cost of a single inference may also increase if model complexity continues to outpace efficiency gains. When considering the electricity mix supplying data centers, rising energy demand is even more concerning.[173]

The notion that AI will solve our climate woes is also contingent on the priorities of AI developers. Although some organizations are applying AI in ways that benefit the environment, most of the industry appears to be more focused on supercharging power and performance. The environmentally optimistic predictions of AI boosters like Sam Altman and Eric Schmidt will only come to pass if high-level decision makers choose to focus on reducing energy demand and its environmental impacts, an outcome that is in tension with the projections of massive electricity demand growth that will outstrip the supply of electricity from the nuclear and renewable energy sectors.

The problem domains to which AI is applied also have profound implications for the impact of AI tools on efforts to address climate change. The oil and gas industry is rapidly adopting AI tools to allow it to extract more oil and gas more cheaply.[174] In turn, this could delay the transition to clean energy sources.

The most important question may not be "to AI or not to AI," but rather, when and how to use AI tools. For example, programmers can save money, time, and energy by opting for coding-specific tools instead of larger general-purpose models. Likewise, fine-tuning (e.g., training a model on a specific dataset to improve performance on a given task) and post-training quantization (e.g., reducing precision of model weights) boost efficiency without seriously affecting response quality.[175] Smaller, more tailored models are often more effective than the largest, most powerful option. However, the lack of consistency across different calculators about which models are more energy efficient than others can make such decisions difficult or impractical.

The implications of AI deployment on the environmental footprint of industry are impossible to predict. It is possible that, like past improvements in industrial productivity, AI deployment will induce companies to build and consume

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173. *See* Guidi et al., *supra* note 15 (concluding US data centers are supplied by electricity that is 48% more carbon intensive than the national average); *see also* Pham Nhat Linh Chi, Le Thi Minh Hang & Nguyen Viet Vuong, *The Negative Impacts of AI on the Environment and Legal Regulation*, 11 INT'L J.L. 124 (2025) (noting that the electricity used to supply data centers currently comes from burning traditional fossil fuels).

174. *See* Sheila Dang & Georgina McCartney, *AI Leading to Faster, Cheaper Oil Production, Executives Say*, REUTERS (Mar. 13, 2025), https://www.reuters.com/business/energy/ceraweek-ai-leading-faster-cheaper-oil-production-executives-say-2025-03-13/.

175. *See generally* Samar Pratap, *The Fine Art of Fine-tuning: A Structured Review of Advanced LLM Fine-tuning Techniques*, 11 NAT. LANG. PROCESSING J. 1 (Jun. 2025) (explaining methods for fine-tuning); *see also* Han Liu, *SEPTQ: A Simple and Effective Post-Training Quantization Paradigm for Large Language Models*, in KDD '25: 31ST ACM SIGKDD CONF. ON KNOWLEDGE DISCOVERY & DATA MINING 812, 813–14 (Ass'n for Computing Machinery) (Jul. 20, 2025) (proposing a more effective method of post-training quantization).

more despite the increased efficiency.[176] There is no 1:1 tradeoff between less efficient human workers and more efficient AI tools because people whose jobs are replaced or improved by AI will continue to emit carbon for some purpose; the emissions may just arise and be attributed differently.[177] As we mentioned above, it is also hard to reconcile projections of surging energy demand attributable to AI and the industry's rapid buildout of energy infrastructure with a vision of the technology as net energy reducing.[178]

Although AI has the potential to reduce energy consumption and accelerate progress towards low-energy solutions, its actual impact remains uncertain. As AI becomes more widespread, providing information about its energy use and environmental effects need not become a barrier to its rapid development and can help reduce costs and environmental impacts.[179] This greater transparency will not arise from federal government pressure in the near term.[180] As the footprint calculator comparison above suggests, though, some AI tools and firms may compare more favorably to others on these metrics. That difference in energy use and environmental impacts, if widely known, could provide these firms with economic incentives to disclose electricity use and environmental impacts, and to market their competitive advantage to companies and individuals who have preferences for efficient, low-environmental impact AI models. Greater transparency is the first step in reducing energy and environmental impacts while harnessing the full potential of AI.

A. Energy Disclosures to Achieve AI's Potential

Understanding the tradeoffs among tools can help users make the most of AI without unnecessarily increasing energy consumption. Without information, though, AI users lack the ability to make smart decisions for themselves, for their companies or other organizations, or for society. Footprint calculators can be especially valuable in this regard because they enable comparison of various options

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176. *See Total Factor Productivity*, BUREAU OF LAB. STAT. (Mar. 21, 2025), https://www.bls.gov/news.release/pdf/prod3.pdf (noting that from 2019-2024, total factor productivity in the private nonfarm business sector grew at an annualized rate of 0.9%, whereas from 2019-2023, capital inputs grew at an annualized rate of 2.6%).

177. It is also possible that an AI tool that is not instructed to prioritize efficiency will embed inefficiencies into its solutions. *See* Md Arman Islam et al., *Evaluating the Energy-Efficiency of the Code Generated by LLMs*, ARXIV (May 23, 2025), https://arxiv.org/html/2505.20324v1 (finding that LLM-generated code is less energy efficient than human-generated code, with human code being on average anywhere from 1.17-2 times more efficient depending on which LLM generated the code).

178. *See* OPENAI, *supra* note 60 (suggesting that the US should add 100GW of new energy capacity each year to meet AI demand).

179. For example, it is possible that AI developers who track and report emissions will find costly algorithmic or hardware inefficiencies. *See* Marinică, *supra* note 31, at 92 ("[W]e appreciate that Al can contribute to the fight against climate change and the consolidation of climate forecasts, but at the same time it also involves costs for the planet, which can be offset or even reduced if a better understanding of the carbon footprint is considered while using Al.").

180. *See supra* notes 21–25 and accompanying text.

in a single space. Rather than defaulting to more expensive flagship models, footprint calculators allow users to compare the costs and qualities of multiple options before selecting the right tool for their task. Doing so requires accurate estimates of energy costs that are not currently available. Model developers can help AI reach its potential by releasing firsthand estimates of relative costs.

Optimizing model decisions should be acceptable to data center operators and model developers, too. Data centers can reduce energy costs while handling the same number of queries if users select more efficient or more flexible options. They also may be able to avoid paying for expensive infrastructure projects designed to handle larger loads. These savings can then be passed on to model developers who agree to design their applications to favor less consumptive versions. Informed users can choose specialized low-cost models that reduce compute, memory, and energy costs for data centers without negatively affecting their experience.

There is a risk that immediate disclosure could incidentally increase overall AI energy demand as users consume more of the relatively less consumptive option. Evidence for this emerges from research on the Jevons Paradox, which suggests that in some cases, efficiency improvements can drive users to consume more, not less, because each marginal unit of consumption costs less.[181] In a market moving as quickly as AI, it is possible that demand will counteract efficiency improvements and increase total energy demand.[182] It is also possible that pro-environmental AI use will crowd out other pro-environmental behaviors if users feel that efficient AI use justifies other, relatively more influential actions.[183] As the 2025 study by Luccioni and colleagues suggests, this means the AI industry should take a more nuanced and wider reaching approach to calculating environmental costs to avoid negative rebound effects.[184]

We also note, though, that the Jevons effect is often far less powerful in the real world than in abstract theory. For example, as energy-efficient LED lighting was starting to become widely available, some analysts predicted that it would lead, through a Jevons effect, to rapidly rising household energy consumption, as the

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181. Mario Giampietro & Kozo Mayumi, *Unraveling the Complexity of the Jevons Paradox: The Link Between Innovation, Efficiency, and Sustainability*, 6 FRONTIERS ENERGY RES. 26, 2 (Apr. 3, 2018) ("Jevons Paradox states that, in the long term, an increase in efficiency in resource use will generate an increase in resource consumption rather than a decrease.").

182. Robert Diab, *Jevons Paradox Makes Regulating AI Sustainability Imperative*, TECH POLICY PRESS (Mar. 11, 2025), https://www.techpolicy.press/jevons-paradox-makes-regulating-ai-sustainability-imperative/ (arguing that the possibility of increases in AI efficiency inducing total consumption reinforces the need for transparency and environmental regulation around AI).

183. *See* Grant. D. Jacobsen, Matthew J. Kotchen & Michael P. Vandenbergh, *The Behavioral Response to Voluntary Provision of an Environmental Public Good: Evidence from Residential Electricity Demand*, 56 EUR. ECON. REV. 946 (Jul. 2012), (exploring whether individuals who voluntarily engage in environmentally behavior do so in order to offset another behavior that is environmentally harmful).

184. Alexandra Sasha Luccioni, Emma Strubell, & Kate Crawford, *From Efficiency Gains to Rebound Effects: The Problem of Jevons' Paradox in AI's Polarized Environmental Debate*, ARXIV, 1 (June 13, 2025), https://arxiv.org/pdf/2501.16548 ("We contend that a narrow focus on direct emissions misrepresents AI's true climate footprint, limiting the scope for meaningful interventions.").

decreasing cost of lighting led people to install more and brighter lights. In fact, the opposite happened, and as more than a billion energy-efficient light bulbs were installed in US homes, per-capita residential electricity consumption dropped considerably.[185]

The wide variations in footprint calculator outputs when evaluating the same queries suggests that users lack access to consistent, reliable information about the energy and environmental impacts of their AI use. Public surveys indicate that protecting the environment is important to a large majority of the American population, including consumers, employees, and managers of companies and other organizations.[186] In the absence of reliable information, AI users cannot act on their preferences to reduce the environmental impacts of their AI use. Public or private standards for AI energy and environmental disclosure could address this problem and enable footprint calculators to provide the public with accurate estimates. Doing so also could align the needs of users, model developers, and data center operators.

CONCLUSION

Ensuring that information is available to the public has been an important foundation for environmental protection in the United States for at least fifty years. Major statutes ranging from the Clean Air Act,[187] the Clean Water Act,[188] and the Toxic Release Inventory provisions of the Emergency Planning and Community Response-to-Know Act[189] all rely on public disclosure of information to enable the public, corporate managers, investors, and government policymakers to make informed decisions about environmental protection. Energy and environmental footprint calculators are a principal means of conveying information about the use of AI. Yet this Article has compared the outputs of four AI-focused footprint calculators and concluded that they produce estimates that vary by as much as 58 times when assessing the same use of AI. This wide disparity in output, despite the same input, indicates that reliable information is not available to the public about the energy and environmental impacts of AI.

This lack of information will not be important if AI energy use does not drive significant growth in electricity demand and environmental impacts, but the

185. Jonathan M. Gilligan & Michael P. Vandenbergh, *A Framework for Assessing the Impact of Private Climate Governance*, 60 ENERGY RSCH. & SOC. SCI. 101400, 2-3 (Feb. 2020) (citing W. Davis, *Evidence of a Decline in Electricity Use by U.S. households*, 37 ECON. BULL. 1098–1105 (2017).

186. Jen Fisher et al., *The Important Role of Leaders in Advancing Human Sustainability*, DELOITTE INSIGHTS (June 18, 2024), https://www.deloitte.com/us/en/insights/topics/talent/workplace-well-being-research-2024.html (93% of executives and 88% of workers believe that the purpose of their company should be to create societal value); NIELSENIQ, SUSTAINABILITY: THE NEW CONSUMER SPENDING OUTLOOK 3 (2022), https://nielseniq.com/wp-content/uploads/sites/4/2022/10/2022-10_ESG_eBook_NIQ_FNL.pdf (76% of consumers value sustainability in their purchases).

187. 42 U.S.C. § 7414 (1970).

188. 33 U.S.C. § 1342 (1972).

189. 42 U.S.C. § 11023 (1986).

available studies suggest that AI is responsible for rapidly growing electricity and environmental impacts. To address this information gap, this Article argues that public or private actors should induce public disclosures from AI firms and data center operators. Governments are unlikely to play this role in the near term, but the analysis in this Article suggests that wide variations in energy consumption likely exist among AI firms, models, and queries, and thus that some firms may have incentives for additional disclosure.[152] In turn, this disclosure can drive changes in the timing, type, and amount of demand in ways that can reduce the energy and environmental impacts of AI. This disclosure can be done without slowing the development of this promising new technology. As its proponents argue, AI has remarkable potential to contribute to economic success and environmental protection, but greater transparency will be necessary to achieve the net contributions of AI.